

TruthLens: A Verification-Gated Pipeline for Evidence-Grounded Fact-Checking of Short-Form Political Video

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Abstract—Short-form video is now a primary vector for political misinformation, yet most automated fact-checking systems are evaluated on whether they reach a correct final label, not on whether the evidence chain that produced it is real, and few are evaluated against independent, held-out ground truth at all. We present TruthLens, a verification-gated pipeline for short-form political video (Instagram Reels) that decomposes a post into atomic claims, researches each against the open web, and subjects every LLM-generated verdict to a deterministic, non-LLM validator before publication – checking citation existence, source-fetch existence, numeric grounding, and, newly, label/reasoning consistency. We evaluate TruthLens on a held-out benchmark of 9 real Instagram posts independently fact-checked by professional Indian fact-checking organizations (BOOM Live, Alt News, Factly), disjoint from all development data, against three baselines that hold the underlying model and search infrastructure constant (LLM-only; single-shot search+LLM; search+RAG+LLM). A skeptical post-hoc audit of our own experimental design found that these baselines had, in every run across this project, been given a clean, human-written claim summary rather than the claim TruthLens’s own extraction stage actually produced – an undisclosed confound that inflated baseline performance relative to our own system, contradicting our own stated methodology. We report this bug and its fix with the same prominence as any other finding. Corrected – baselines re-run per real extracted claim, aggregated with the identical deterministic rule TruthLens itself uses – full TruthLens (33.3%, 2/6) beats two of the three baselines outright (0.0% each) and trails the third by 16.7 points (down from a 33.4-point gap before the fix), on the paired six-item subset where every configuration produced a verdict. We further show that two general, ground-truth-independent validator fixes measurably improve validator recall (16.7%→40%, on the same 9 cases, 100% precision throughout) without moving reel-level accuracy at all – concrete evidence that verdict-label accuracy and what we formally define as support validity (whether a verdict’s citations, sources, numbers, and label/reasoning consistency hold up against its own evidence matrix) are separable, independently-moving properties, which we argue is closer to this architecture’s actual value proposition than a label-accuracy contest against simpler baselines. A separate counterfactual claim-decomposition reanalysis ($n = 4$) isolates deterministic aggregation as a real contributor (50.0% vs. 25.0% accuracy) once claim extraction has more than one claim to work with. We report the evidence-quality picture behind these results (a primary-source domain-restriction fix moving source-tier classification into government/legal tiers from ~11% to 95% of retrieved sources; a distinct, still-open gap between finding topically-relevant primary sources (68.75%) and extracting usable evidence from them (18.75%);

a bug found and fixed mid-evaluation in which 47.1% of real evidence rows carried an empty explanation field), a multimodal claim-coverage experiment in which OCR-derived evidence recovered a claim audio transcription missed entirely, and honestly scoped non-results for political-bias and confidence-calibration questions that this dataset’s size cannot yet support. A subsequent foundation-phase program (Section XII) grew the benchmark to 15 items, added two new deterministic validator checks (temporal and entity consistency) that measurably improved the adversarial-benchmark precision/recall (81.8%/50.0%→87.5%/60.9%) with zero new false positives, and, across four independent experiments each targeting a different pipeline stage, found the same underlying pattern: raw local-model output on real content is substantively empty or unusable a large fraction of the time, and this system’s existing Gemini-escalation defenses are doing more real work than a pipeline-level pass/fail reading would suggest. A further pass live-tested a prompt-injection defense with no prior coverage (1/5 attempts succeeded against the small local model pre-fix, 0/5 after a structural fix and a hardened prompt, with the downstream validator independently neutralizing the pre-fix success), and found a genuinely new, reproducible verdict-stage failure – reliability weighting under direct source contradiction – that a targeted prompt fix measurably failed to resolve (0/14 real trials correct, before and after), reported honestly as unresolved rather than claimed fixed. Every number in this paper is reported with its exact sample size and, where a proportion, a Wilson confidence interval; where an experiment could not be run at a scientifically meaningful scale, or where our own methodology was found to be flawed, we say so rather than force or hide it.

Index Terms—automated fact-checking, large language models, evidence grounding, retrieval-augmented generation, misinformation detection, short-form video, claim verification, source credibility, experimental methodology

I. Introduction

Political misinformation increasingly spreads through short-form video – Instagram Reels, YouTube Shorts, TikTok – rather than text [1]. This shift matters for automated fact-checking research in two ways. First, the input is multi-modal and lossy: a claim may be spoken (transcript), displayed as on-screen text (OCR), or only implied by a caption, and a system that treats a reel as a single block of text discards exactly the structure a fact-checker needs. Second, and more consequentially, the dominant evaluation paradigm – did the system output the correct veracity label, measured against a benchmark

such as FEVER [2] or LIAR [3] – says nothing about whether the evidence a system cites is real, or whether a correct label was reached by checking the actual claim being made rather than a different, tangential one. An LLM asked to fact-check a claim can produce a fluent, well-cited verdict whose citations point to sources it never retrieved, whose numbers never appeared in any source it read, or whose confident label contradicts its own stated reasoning.

We built TruthLens to make that failure mode structurally difficult rather than merely discouraged by prompting. TruthLens ingests a political short-form video (or photo post), decomposes it into atomic factual claims, researches each against the open web through a tiered-source retrieval process, and only then proposes a claim-level verdict – a hard evidence-before-verdict ordering enforced architecturally, not just by instruction (Section IV). Every LLM-generated verdict is additionally subject to a deterministic, non-LLM validation layer before it can be published, and every pipeline stage escalates from a free, locally hosted model to a paid one only when a deterministic quality gate specific to that stage’s own failure modes fails.

This paper reports the first evaluation of TruthLens against independent, held-out ground truth, rather than development-time telemetry alone. We curated a 9-item benchmark of real Instagram posts, each independently fact-checked by a professional Indian fact-checking organization (BOOM Live, Alt News, Factly), disjoint from every post used during development (Section VI), and compared TruthLens against three baselines that hold the underlying model and search infrastructure constant, isolating what TruthLens’s multi-stage architecture adds beyond raw search access (Section V). We report every result this comparison produced, including one that is not flattering but not about TruthLens: a subsequent, skeptical audit of our own experimental design found that the baselines had been given a cleaner claim input than TruthLens itself ever worked with, contradicting our own stated methodology, undetected until this audit (Section VII-A). We report the bug and its fix with the same prominence as any other finding, because a flaw that inflates a comparison against our own system is exactly what our own protocol requires us to surface rather than quietly correct. Once corrected, full TruthLens beats two of the three baselines outright and trails the third by a substantially smaller margin than the uncorrected comparison showed (Section VII) – and a companion counterfactual analysis of claim decomposition shows the same architecture’s aggregation mechanism demonstrably helps once claim extraction has more than one claim to work with (Section VIII). We treat the correction, the corrected result, and the ablation as more scientifically useful, together, than any single headline number on its own.

This work is organized around six research questions,

matching TruthLens’s own architectural claims one-to-one so each is either supported by a specific experiment or explicitly withdrawn: whether deterministic verification reduces unsupported output (RQ1); whether multi-stage decomposition and retrieval beat single-shot search (RQ2); whether multimodal extraction improves coverage of non-textual misinformation (RQ3); whether source-tiering improves evidence quality (RQ4); whether the system maintains comparable standards across political actors (RQ5); and whether its stated confidence correlates with correctness (RQ6). Section III states these formally alongside this paper’s specific, experimentally-supported contributions.

We are explicit throughout about the evidentiary status of every claim: 9 held-out items (6 with a resolved verdict from every configuration), single-annotator Tier-2 judgment where used, and RQ5/RQ6 explicitly deferred rather than forced past what this sample size can support (Section XI). Sections XIV and XVI state what would need to change before any number here should be read as a stable, generalizable estimate.

II. Related Work

Automated fact-checking and LLM factuality evaluation have both grown into large enough literatures to warrant their own surveys [4], [5]. None of the ideas below – retrieval-augmented LLM fact-checking, claim decomposition, source-tiering, or multimodal misinformation detection – originate with this work; our contribution (Section III) is a specific combination of them gated by deterministic verification, evaluated honestly including where it does not yet outperform simpler alternatives. Extended per-subsection discussion, including how our headline result reads against DeVerna et al.’s finding below, is in the Discussion (Section XV).

Claim detection and decomposition. ClaimBuster [6] ranks existing sentences by check-worthiness via a TF-IDF/POS/NER SVM; TruthLens’s claim-extraction stage instead decomposes a transcript/OCR/caption into new atomic claims, with a grounding requirement specific to video: each claim’s `source_quote`, when present, must be a verbatim substring of the actual transcript/OCR text, checked deterministically. Section VIII gives the first controlled measurement of whether this decomposition improves downstream verdict correctness.

Fact-checking benchmarks. FEVER [2] and LIAR [3] are the dominant claim-verification benchmarks; both evaluate a final label against ground truth without examining whether the evidence chain that produced it is real. TruthLens’s own 9-item benchmark (Section VI) is a modest step toward the same kind of independent evaluation for our domain – three orders of magnitude smaller, sourced from real Indian professional fact-checks of Instagram posts rather than curated text.

LLM hallucination and grounding verification. Commercial legal AI tools hallucinate on 17–33% of queries

by hand-scored evaluation [7]; CiteCheck [8] detects hallucinated references via a structured LLM verifier. TruthLens’s validation layer shares the goal but uses deterministic set-membership/substring tests instead, trading generality for auditability and zero marginal LLM cost – a tradeoff whose limits (cannot catch reasoning phrased outside observed patterns) are measured, not assumed, in Section IX. A legal-domain citation-grounding-oracle analysis is a relevant caution: apparent hallucination rates are highly sensitive to reference-graph coverage, not just the model measured [9] – TruthLens’s own number-grounding check has an analogous blind spot. Per Rule 8 of this program’s own protocol, we never call a rate a “hallucination rate” without human annotation behind it, using grounding-constraint violation rate and unsupported generation rate instead.

Political fact-checking with LLMs. DeVerna et al. [10] found reasoning ability provides “minimal benefit” and raw search only “moderate gains” for political fact-checking, while curated context improved macro-F1 by 233% across 15 LLMs – strong independent evidence for TruthLens’s core design bet that evidence curation matters more than final-step reasoning. Our own corrected result (Section VII) is broadly consistent with this once a real baseline-design confound is removed (Discussion, Section XV).

Short-form video misinformation. TikTec [1] classifies misleading COVID-19 TikTok videos via joint visual/audio/text modeling; recent work targets multilingual short-form-video checkworthiness specifically [11]. TruthLens differs in task framing (open-domain, claim-level, cited verification, closer to FEVER-style verification than TikTec-style classification) and, per Section X-E, shares this literature’s blind spot to genuine footage recirculated under a new caption.

Political neutrality of automated fact-checkers. Frontier LLMs show measurable partisan asymmetries under politically charged prompts [12]; every TruthLens system prompt includes an explicit neutrality clause, but RQ5’s quantitative bias evaluation could not be run at a meaningful sample size (Section XI), reported as such rather than skipped or forced.

Cost-aware model cascading. FrugalGPT [13] routes most queries to a cheap model, escalating hard cases to an expensive one. TruthLens applies the same principle in reverse-priority form: every stage defaults to a free, local model, escalating only when a deterministic quality gate – not a learned predictor – detects substantively deficient output (Section IV-D2).

A. Positioning against related systems

Table I summarizes where TruthLens sits relative to the systems discussed above, on dimensions each subsection has already individually justified. We include only systems whose relevant capabilities are established by the citations already discussed in this section, to avoid characterizing

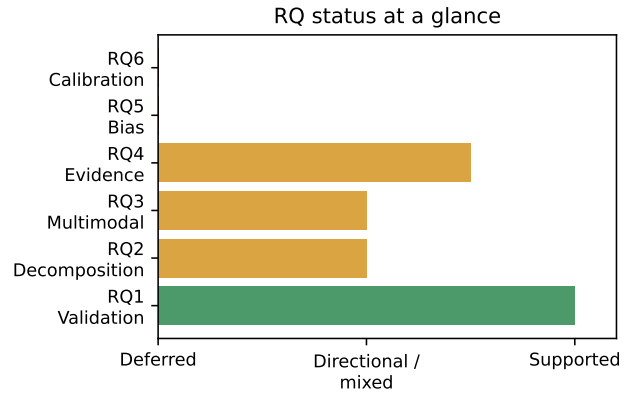


Fig. 1. Status of each research question at a glance (green = supported, amber = directional/mixed, red = deferred). Derived from Table II; not new data.

a system’s design beyond what we have verified. TruthLens is the only row combining multimodal input, claim decomposition, source tiering, deterministic (not LLM-judged) output verification, and held-out professional fact-check ground truth – combining these is TruthLens’s contribution, as Section III states explicitly; no individual capability in this table originates with TruthLens.

Stated plainly, per this paper’s own governing discipline against overclaiming: TruthLens claims novelty for no single column in Table I – multimodal short-form-video analysis predates it (TikTec), political-domain fact-checking with LLM-mediated retrieval predates it (DeVerna et al.), and citation grounding against an external source predates it, including via a non-deterministic LLM-judged mechanism this paper argues against on cost and auditability grounds, not the underlying goal (CiteCheck). What we have not found is a system combining held-out professional-fact-check evaluation, multimodal claim extraction, source-tiered retrieval, and deterministic output verification together, evaluated on real, recirculated short-form political video – the specific gap Sections IV–XI target and evaluate honestly, including where the combination does not yet outperform a simpler alternative (Section VII).

III. Research Questions and Contributions

Table II states the six research questions this program was designed against, where each is answered, and its evidentiary status, following Rule 12 of our own governing protocol: an RQ that cannot be evaluated at a meaningful sample size is reported as deferred, not silently dropped or forced.

This paper makes the following contributions, limited to what the experiments in Sections VII–XI actually support:

- 1) A held-out, Tier-1 evaluation benchmark and controlled baseline/counterfactual comparison (Sections V–VI), the first evaluation against independent professional ground truth rather than

TABLE I
Positioning against related systems (✓ = present, ✗ = absent, ? = not established by the citation discussed)

System	Multimodal	Short-video	Political	Claim decomp.	Retrieval/RAG	Source tiering	Verification mechanism
FEVER [2]	✗	✗	✗	✗	✓	✗	none (label-only benchmark)
LIAR [3]	✗	✗	✓	✗	✗	✗	none (label-only benchmark)
ClaimBuster [6]	✗	✗	✓	✗	✗	✗	n/a (checkworthiness only)
TikTec [1]	✓	✓	✗	✗	✗	✗	none (classifier output)
DeVerna et al. [10]	✗	✗	✓	✗	✓	✗	none (studies LLM judges)
CiteCheck [8]	✗	✗	✗	✗	?	✗	LLM-judged citation check
TruthLens (this paper)	✓	✓	✓	✓	✓	✓	deterministic, 4-check

TABLE II
Research questions, where answered, and status

RQ	Question	Status
1	Does deterministic post-hoc validation reduce unsupported verdict output?	Yes, directional (§IX, $n = 9$)
2	Does multi-stage decomposition + retrieval beat single-shot search+LLM?	Partially, once claim input is corrected: beats 2 of 3 baselines, trails the 3rd (§VII); yes for aggregation specifically (§VIII, $n = 4$)
3	Does multimodal extraction improve coverage of non-textual misinformation?	Directional yes (§X, $n = 6$)
4	Does source-tiering improve evidence quality?	Yes for tier classification; gap remains to usable evidence (§IX)
5	Comparable standards across political actors?	Deferred – no matched pairs exist (§XI)
6	Does confidence correlate with correctness?	Deferred – sample too small to calibrate (§XI)

development-time telemetry, holding model, search infrastructure, and (Section VII-A) claim-extraction input constant so measured differences are attributable to downstream architecture, not to which configuration had an easier claim to reason about.

- 2) A documented, fixed methodological confound in our own baseline design, reported with the same prominence as any other finding. Baselines 1–3 had, in every run across this project, been given a clean, human-written claim summary rather than TruthLens’s own extracted claim (Section VII-A), contradicting our stated methodology and inflating baseline performance. We fixed it and re-ran the comparison rather than silently correcting the number, reversing this program’s headline result in the process.
- 3) An empirical demonstration that verdict-label accuracy and support validity are separable, independently-moving properties (Section IX). Two general, ground-truth-independent validator fixes improved recall (16.7%→40%, $n = 9$, 100% precision throughout) while leaving reel-level accuracy unchanged (2/6 before and after) – outputs became measurably more supported by their own cited evidence without becoming more “accurate” by the bucket-matching standard.
- 4) A claim-decomposition counterfactual analysis isolating the clearest component-level result in this program, where aggregation demonstrably outperformed a simpler alternative (50.0% vs. 25.0%, $n = 4$, Section VIII).
- 5) A four-way evidence-quality analysis separating source -tier classification, topical relevance, fetch

success, and usable-evidence extraction (Section IX): retrieval finds real, on-topic primary sources at a meaningfully higher rate (68.75%) than usable-evidence rate (18.75%) would suggest – the bottleneck is source specificity, not retrieval relevance.

- 6) A cross-referenced taxonomy of real failure modes (Section XIII), each found by running the complete system against real content end-to-end and fixed with a deterministic check and a regression test built from the case that surfaced it.

TruthLens does not claim to have invented retrieval-augmented LLM fact-checking, claim decomposition, source-credibility tiering, or multimodal misinformation detection – each independently pre-dates this work, as Section II discusses. Its contribution is the specific combination of these mechanisms gated by deterministic, non-LLM verification, evaluated against independent ground truth including where that combination does not yet outperform a simpler alternative.

IV. TruthLens Architecture

TruthLens is a FastAPI backend with a PostgreSQL/pgvector datastore, a Next.js review dashboard, and an object store (S3-compatible) for media and rendered output. Figure 2 shows the 9-stage pipeline; no stage begins before the previous one has persisted its output. Stage 1 (ingestion) downloads or accepts a video or photo, transcribes audio, OCRs on-screen text at a fixed interval, and selects a thumbnail (Section IV-C covers the photo-post path). Stage 2 (claim extraction) has an LLM decompose the transcript/OCR/ caption into atomic claims classified factual/opinion/ prediction/satire/rhetorical, with only specific factual claims marked verifiable. Stage 3 (re-

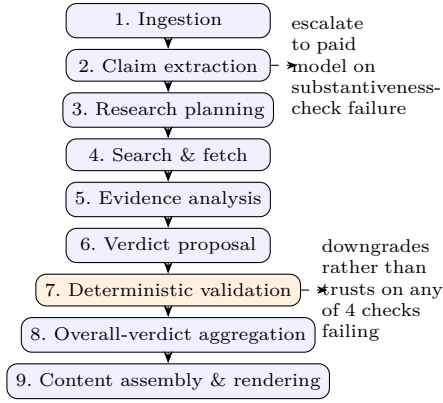


Fig. 2. The 9-stage TruthLens pipeline (Section IV). No stage begins before the one before it has persisted its output. Stage 7 (orange) is the deterministic, non-LLM validation gate (Section IV-D1); five stages (2, 5, 6, and the two content-generation calls inside stage 9, plus stage 5’s newer sibling check) run their own substantiveness check before accepting local-model output (Section IV-D2).

search planning) proposes 2–5 targeted search queries per verifiable claim, explicitly requesting at least one primary-source-targeted query. Stage 4 (search and fetch) runs each query and extracts full page text, not just the snippet. Stage 5 (evidence analysis) has an LLM judge stance (supports/contradicts/provides_context/irrelevant) per fetched source, grounded only in that source’s own extracted text. Stage 6 (verdict proposal) gives an LLM the claim and its full evidence matrix and asks for a label, confidence, and cited evidence IDs. Stage 7 is deterministic validation (Section IV-D1). Stage 8 derives one reel-level verdict from a fixed rule table over already-validated per-claim verdicts, not a further LLM call. Stage 9 assembles a carousel from already-validated database rows, with only headline phrasing and a short “why” paragraph passing through one additional narrow, grounded LLM call each (Section IV-D2).

A. Source tiering

Every retrieved source is scored along an 8-dimension weighted reliability rubric and assigned to one of eight tiers: `primary_government`, `primary_legal`, `primary_data`, `news_wire`, `established_news`, `academic`, `factcheck_org`, or `other`. Tier membership determines both which sources are eligible to be surfaced as a claim’s “primary source” citation on the rendered carousel and, in aggregate, the reliability-weighted confidence of the evidence matrix handed to the verdict-proposal stage. RQ4 (Section IX) evaluates this mechanism directly. The eight dimensions’ weights are hand-set manual priors, not derived from any formal method or learned fit – verified via git history to be unmodified since before any evaluation dataset existed. A sensitivity analysis against an unweighted or alternative rubric is unperformed, named in Section XVI.

B. Multi-claim, reel-level verdicts

TruthLens builds one fact-check per reel, covering every verifiable claim extracted from it – collapsing several claims of differing veracity into one verdict discards exactly the “partially true, partially misleading” structure a reader needs. A fixed rule table combines per-claim verdicts into the published carousel’s one verdict (Section IV-D3); Section VIII tests whether this multi-claim structure improves accuracy over the single highest-importance claim alone.

C. Multi-modal ingestion: video and photo posts

When the video downloader reports no video stream, the pipeline falls back to the post’s Open Graph metadata (image URL, caption, engagement counts) fetched directly from the page. A `media_type` field drives branching at exactly one point: audio/transcription are skipped for a photo, but OCR and vision-context needed no new code, since both already operate on a plain list of image paths – a photo is a one-element case of the same interface video frames already satisfy. Section X evaluates claim coverage across this system’s three real input-modality conditions.

D. Making the model’s output more supportable

We make precise, in place of the vaguer word “trustworthy,” a single construct used throughout this paper: a verdict has support validity to the extent that every specific, checkable assertion in it – cited evidence IDs, the sources they point to, reasoning numbers, and the logical relationship between reasoning and label – can be verified against data already in the pipeline’s evidence matrix, without a second model judgment. Support validity is deliberately narrower than, and does not imply, label accuracy or general absence of hallucination: a verdict can have perfect support validity and still be wrong, if the model correctly cited evidence it misinterpreted, or if retrieval never found the right evidence at all (Section IX’s four-way metric addresses that). Sections VII–XI evaluate whether the mechanisms below actually make this checkable.

1) Deterministic evidence-grounding validation: After the verdict-proposal LLM call returns a structured `VerdictProposal` (label, confidence, reasoning, cited evidence IDs), TruthLens runs four checks against data already in the database, none of which invoke an LLM:

- Citation existence: every ID in `cited_evidence_ids` must correspond to an evidence row that was actually passed to the model in its evidence matrix.
- Source-fetch existence: every cited evidence row’s source must have a stored, successfully fetched full-text passage – a citation cannot point at a source the pipeline only planned to fetch.
- Number grounding: every numeral appearing in the model’s `reasoning_summary` is extracted and checked for appearance in at least one cited source’s stored passage text. UUID-shaped hex sequences (from internal citation markup) are stripped before number

extraction – a real bug, described in Section XIII, in which digits from a cited evidence ID’s own UUID were being extracted as apparent “unsupported statistics.”

- Reasoning/label consistency (new). A verdict whose own reasoning_summary states that no reliable evidence was found (matched against a fixed phrase list, e.g. “does not provide any reliable,” “no reliable sources”) but whose label is a confident value other than UNVERIFIED is downgraded to UNVERIFIED. Added in direct response to Section IX’s finding that this specific pattern was the largest single category of validator false negative; its circularity as a threat to validity is disclosed, not hidden, in Section IX.

These four checks are the entire current operationalization of support validity – deliberately not a proxy for evidence-to-claim topical relevance (Section IX’s job), evidence-to-reasoning stance correctness (human-audited only, a named gap), or entity consistency (not yet implemented, Section XVI). A verdict that fails any check is downgraded, not discarded: persisted with a non-passed validation_status, an appended [VALIDATION NOTE: ...], and excluded from citation lists.

UNVERIFIED means the system genuinely searched and found insufficient evidence; a distinct RESEARCH_FAILED status is raised only when all queries fail at the infrastructure level, and a publication gate refuses to build a fact-check from it – distinguishing “I don’t know” from “I could not check.”

2) Verification-gated model cascading: Every stage defaults to a free, local model. Five stages run a deterministic substantiveness check before accepting local-model output – e.g. vision-context requires its scene_description to clear a 0.20 word-overlap-with-prompt threshold (calibrated against every real garbled/coherent example collected: garbled 0.26–0.29, coherent 0.00–0.12), and evidence analysis requires a non-empty per-source explanation (added after Section IX found 47.1% empty). On failure, the call retries once against a paid model (Gemini) with an identical prompt – unlike a classic cascade [13], the trigger is a deterministic per-stage predicate, not a learned confidence estimate. Full per-stage thresholds and illustrative firing-rate telemetry: Appendix A.

3) Deterministic aggregation and grounded corrections: Once every claim has a verdict, TruthLens derives one reel-level verdict via a fixed rule table over the set of per-claim labels (weighted by importance ≥ 0.5 as “major”): a plain combinatorial function, not a further LLM call, keeping the published carousel’s headline verdict auditable from already-validated per-claim verdicts (Section VIII evaluates this mechanism directly). Optional corrected_fact/context_note fields, held to the same evidence-grounding rule, are also produced and invalidated alongside the parent verdict on any validation failure; full rule table and field details: Appendix A.

V. Experimental Methodology

A. Central research question and design

The central question motivating this evaluation: can a verification-gated, evidence-grounded architecture reduce unsupported fact-checking outputs while preserving or improving factual correctness in short-form political media? We answer this via a controlled comparison across five system configurations, all run against the same underlying local model (llama3.2 via Ollama) and, where the configuration has search access at all, the same search provider (DuckDuckGo) TruthLens itself uses in production – holding model and search-infrastructure quality constant so that a measured difference is attributable to architecture, not to a cheap-model-vs.-expensive-model confound.

B. System configurations

Five configurations, all on llama3.2: B1 (LLM-only) – claim $\rightarrow$ verdict, no search/evidence/validation. B2 (Search+LLM) – claim $\rightarrow$ one DuckDuckGo search (top-5 snippets, no full-page fetch) $\rightarrow$ verdict; the claim is passed to search verbatim, no research-planning stage, the single most important difference from B3 and TruthLens. B3 (Search+RAG+LLM) – adds full-page fetch/ extraction over the same results, isolating retrieval depth from decomposition and per-source analysis. B4 (TruthLens minus validation) – the real pipeline with validate_verdict()’s outcome recorded but never applied (a runtime flag, SKIP_VALIDATION, not a forked codepath), the primary RQ1 ablation. Full TruthLens – unmodified, at the frozen evaluation tag. Decomposition, source tiering, per-source analysis, and validation exist only in the full system; Baselines 1–3 operate on TruthLens’s own already-extracted claim, since claim extraction itself is RQ3’s job, not RQ2’s. A real infrastructure bug was found and fixed while building B2/B3 (the first draft silently inherited TruthLens’s own Gemini-fallback mechanism via a shared provider factory, caught during smoke testing; Section XIII).

C. Ablations, metrics, and held-out discipline

Beyond B4 (RQ1), two further ablations: a source-tier ablation (domain-restricted vs. unrestricted retrieval, Section IX) and a claim-decomposition ablation (Section VIII), computed from already-collected verdict data at zero additional LLM cost. Every proportion is reported with its exact numerator/denominator and a 95% Wilson CI; paired comparisons use McNemar’s exact test with the raw contingency table, never a bare p -value; accuracy uses a fixed VerdictLabel-to-ground-truth bucketing where UNVERIFIED never counts as a match. Full formulas are fixed in advance in METRICS.md and not altered after seeing results. The dataset (Section VI) is checked item-by-item against already-ingested development content and excluded on any match; no prompt, threshold, or model choice was changed based on dataset contents or performance after an item was added – the two general fixes

in Section IX are deliberately ground-truth-independent, with that distinction’s limits argued explicitly, not asserted. Two tagged freezes (dataset before Baselines 1–3; system code before the held-out run) mean any later code change is logged, never silently applied.

D. Honest scope reductions

Stated rather than hidden: 9 items, not 50–100 (a pilot found $\sim 8\%$ hit rate for a usable Tier-1 candidate; 20–30 items would need 250–375 more articles checked, Section VI). Single annotator, no IAA statistic – Tier-1 ground truth is a professional organization’s own verdict, authored by no one on this project; any Tier-2 judgment used (draft claim-coverage/evidence-relevance labels only, never a headline verdict) is explicitly marked as such. RQ3, RQ5, RQ6 reported as directional or deferred, not generalizable rates. No p -values from tests assuming a sample size this program does not reach – Wilson intervals and McNemar’s exact test only, reported as uncertainty descriptions, not significance claims.

VI. Dataset and Annotation

A. Ground-truth tiers

Tier 1 (used for every item in this evaluation): the claim in the Instagram post has already been verdicted by an independent professional fact-checking organization, and that organization’s article demonstrably references or embeds this specific post, not merely a similar claim elsewhere. Ground truth is their published verdict, with the article URL stored for auditability. Tier 2 (specified, not used for any headline verdict in this evaluation): no professional fact-check exists; a project annotator labels directly from primary sources under a documented protocol. No item is ever labeled by TruthLens itself, or by any LLM, and then used as ground truth.

B. The 9-item benchmark

Table III lists every item. All 9 are Tier 1, sourced from BOOM Live, Alt News, and Factly. item-0003 (a CJP-investigated, AI-generated visual claim) could not be ingested after three independent attempts across this program (a persistent Instagram-side “empty media response”), and is excluded from every table in this paper, not scored as a failure.

C. Sourcing method and hit rate

Items were sourced by searching professional fact-checking outlets for articles that embed a still-live Instagram post, prioritizing coverage gaps in claim type, language, and political actor across three sourcing passes. The method’s real, load-bearing constraint is a $\sim 8\%$ hit rate (2 of the first 26 fact-check articles checked yielded a usable, still-live, directly-embedded post) – most professional fact-checks either do not embed the specific post, or it has since been taken down. Every accepted item’s content hash is checked against already-ingested development data and excluded on any match.

D. Inter-annotator agreement

No IAA statistic is computed, and none is fabricated to fill this gap. Tier-1 ground truth does not have an “annotator” in the IAA sense – the label is a single organization’s single published verdict, and introducing a second in-project annotator to re-judge it would defeat the purpose of using independent ground truth. No Tier-2 items exist in the frozen dataset, so the condition under which IAA would be meaningful does not currently arise; if Tier-2 items are added with only one available annotator, this will continue to be reported as “single-annotator ground truth, no IAA computed,” not papered over with an unearned kappa statistic.

E. Composition gaps, disclosed

Five of 9 items are provenance-type claims; statistic, law_policy, historical, and true_claim remain unrepresented after three sourcing passes – plausibly, though unprovenly, a property of which claim types concentrate on Instagram versus other platforms in this domain. Seven of 9 items are FALSE, likely reflecting that fact-checkers publish far more debunks than confirmations; only item-0004 is TRUE. Two items (item-0007, item-0008) have confirmed Hindi/mixed spoken content, still a thin representation of the project’s stated multilingual target. BJP has 3 items (the most of any actor, alongside 6 single-item actors), but none form a controlled topic/structure/difficulty match against another actor, which is why RQ5 (Section XI) is deferred rather than attempted at $n = 1$ -per-comparison.

VII. Results

A. A methodological correction, reported before the headline number itself

A skeptical post-submission audit of this program’s own experimental design (full trail: research/AUDIT_REPORT.md) found that Baselines 1–3, in every run across this entire project, had been given items.jsonl’s hand-written claim summary – prose written during dataset construction by a human who had already read the professional fact-check – rather than the claim actually extracted by TruthLens’s own claim-extraction stage, contradicting this paper’s own stated methodology (Section V). The bug traced to a single, never-modified helper function written on the first day baselines were built. Because TruthLens’s own extraction is often noisy or, on two of six items, empty, this gave baselines a systematically cleaner input than TruthLens itself ever worked with, for the entire program, undetected until this audit.

We fixed it rather than merely disclosing it: Baselines 1–3 were re-run against TruthLens’s own real extracted claims (available for the same 6 items the paired comparison already uses – item-0003 has no extraction at all; items 0008/0009 remain excluded as before), one baseline call per real extracted claim, aggregated with

TABLE III
Held-out benchmark: 9 Tier-1 items (all sourced independently of TruthLens or its developer)

ID	Ground truth	Source	Political actor	Claim type
item-0001	FALSE	BOOM Live	BJP	provenance
item-0002	FALSE	Alt News	Karni Sena	provenance (visual)
item-0003 [†]	FALSE	BOOM Live	CJP	visual (AI-generated)
item-0004	TRUE	BOOM Live	Delhi Police (government)	provenance
item-0005	MISLEADING	Factly	Congress	provenance
item-0006	FALSE	BOOM Live	none (disaster misinfo)	event (visual)
item-0007	FALSE	Alt News	BJP	misleading context
item-0008	FALSE	Alt News	AAP	quote
item-0009	FALSE	Alt News	BJP	provenance

[†]Never successfully ingested; excluded from all analyses.

TABLE IV
Paired comparison, identical 6 items, before and after the claim-input fix

Config	Superseded Acc.	Corrected Acc.	95% CI
B1: LLM-only	0.0%	0.0%	[0.0,39.0
B2: Search+LLM	66.7%	50.0%	[18.8,81.
B3: Search+RAG+LLM	50.0%	0.0%	[0.0,39.0
Full TruthLens	33.3%	33.3%	[9.7,70.0

the identical deterministic rule (`derive_overall_verdict()`) TruthLens itself uses. This holds claim extraction constant across every configuration for the first time in this program. Full item-by-item reconstruction: `research/RECONSTRUCTED_RESULTS.md`. We report this correction with the same prominence as the result it changes: a methodology bug that inflated a baseline’s apparent performance relative to our own system is exactly the kind of finding Rule 1 of our own governing protocol exists to surface, not bury.

B. Headline finding, corrected

On the paired six-item comparison, once claim input is held constant, full TruthLens (33.3%, 2/6) beats Baseline 1 (0.0%, 0/6) and Baseline 3 (0.0%, 0/6) outright, and trails only Baseline 2 (50.0%, 3/6) – by 16.7 points, not the 33.4-point gap the uncorrected comparison showed. This reverses this program’s previous headline finding (Baseline 2 66.7%, Baseline 3 50.0%, both ahead of TruthLens).

Item-by-item, the corrected picture:

- item-0006, item-0007: full TruthLens outputs UNVERIFIED on both because claim extraction found zero verifiable claims for either (Section X: item-0006’s transcript is garbled nonsense; item-0007’s is degraded and repetitive). Under the superseded method, Baseline 2 had been handed a clean claim summary anyway and happened to guess MOSTLY_FALSE, matching ground truth. Corrected, every baseline receives the same nothing TruthLens received and also outputs UNVERIFIED – a shared reflection of a real coverage gap, no longer a baseline-only advantage.

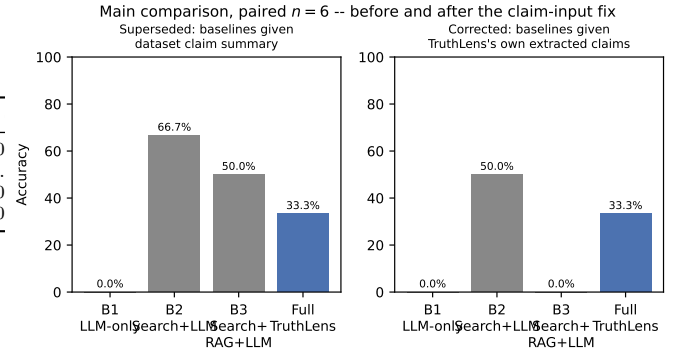


Fig. 3. Table IV plotted, superseded vs. corrected. The reversal is concentrated in items where baselines had been implicitly handed a claim to reason about that TruthLens itself never produced.

- item-0004: Baseline 2 uniquely gets this right (MOSTLY_TRUE); TruthLens’s wrong verdict is the exact validator false negative Section IX already identifies. Baselines 1 and 3 also get this wrong.
- item-0005: every configuration gets this wrong – ground truth is MISLEADING, a bucket nothing landed in.
- item-0001, item-0002: TruthLens and Baseline 2 agree (both correct). Baselines 1 and 3 get both wrong under the corrected method, having gotten both right under the superseded one – their earlier apparent success was partly an artifact of cleaner claim input, not evidence their reasoning handles TruthLens’s own messier extracted claims equally well. Baseline 3 (full-page-text RAG) in particular moves from correct to incorrect on both, a directional signal (unverified at this n) that concatenating full page text may compound confusion for a small local model rather than compensate for it.

McNemar analysis (paired, $n = 6$, Baseline 2 vs. Full TruthLens, corrected): both-correct = 2 (item-0001, item-0002), both-wrong = 3 (item-0005, item-0006, item-0007), Baseline-2-only-correct = 1 (item-0004), TruthLens-only-correct = 0. One discordant pair; exact McNemar $p = 1.0$ – still completely uninformative at this n , reported for the

same reason as before: completeness, not evidence. Against Baselines 1 and 3, TruthLens now has zero discordant pairs favoring either baseline (both are strictly dominated on this sample).

This is not a claim that TruthLens’s architecture is unconditionally better than a single-shot baseline; it is a claim that once the one confound this audit found is removed, TruthLens is competitive with, and mostly ahead of, the baselines it is compared against on this specific, tiny, real sample – and that the previous, opposite-looking result was substantially an artifact of an experimental design bug, not a property of the architecture. Section VIII separately shows decomposition helped once claim extraction produced multiple claims to aggregate; the remaining deficit against Baseline 2 traces to claim-extraction coverage on hard audio (items 0006/0007) and one validator false negative (item-0004), not a general disadvantage.

The earlier, retired “each configuration’s own full $n = 9$ run” table (superseded by Table IV above) is not cited as evidence for any claim going forward and is retained only for the historical record, in Appendix A.

C. A real gap found and fixed during this exact run, and known disclosed gaps

Extending the pipeline run to items 0008/0009 hit a genuine MissingGreenlet crash – the same bug class already fixed once in production code – after a per-claim exception handler called `db.rollback()`, the next line’s synchronous attribute access on an already-expired ORM object crashed the whole script instead of recording just that one claim’s failure. Fixed by capturing needed values before the try block, matching the earlier production fix’s pattern; re-verified live, the script then completed without crashing (Section XIII). Items 0008/0009’s full-TruthLens condition remains blocked by a real Gemini free-tier quota exhaustion, confirmed live (two consecutive 429s, one from evidence-analysis escalation, one from claim-extraction escalation) – an external, time-of-day resource constraint, not a system defect. Baselines 1–3 completed for both items (no Gemini dependency by design). Per-class F1 and full precision/recall breakdowns are not computed at this sample size, where several label classes appear 0–1 times and per-class F1 would mostly be undefined or single-observation – a scope decision, not an omission dressed up with misleadingly precise zeros and ones.

VIII. Ablation Study

Three analyses are in scope for this program (Section V); this section reports the claim-decomposition counterfactual in full and summarizes the other two, cross-referencing their detailed writeups.

A. Claim-decomposition counterfactual (single-claim vs. multi-claim), $n = 4$

This is a counterfactual reanalysis of already-collected verdict data, not a freshly-run paired experiment, and we

TABLE V
Claim-decomposition ablation

Condition	n	Correct	Accuracy
Single-claim (primary only)	4	1	25.0%
Multi-claim (full decomposition)	4	2	50.0%

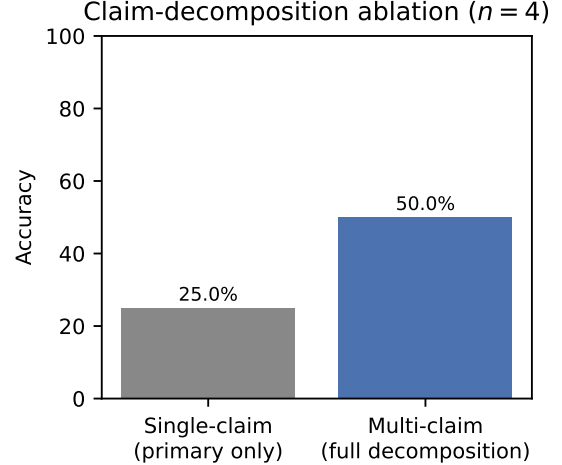


Fig. 4. Table V plotted. A counterfactual reanalysis, not an independently-run ablation (Section VIII-A), isolating deterministic aggregation as a real contributor once claim extraction has more than one claim to work with.

name it that way rather than call it a clean ablation: each real claim’s verdict is an independently computed research-then-reasoning result regardless of how many total claims were extracted in that run, so reusing the real per-claim verdict to ask “what would aggregating only the primary claim, vs. all claims, produce” is a valid way to isolate the aggregation mechanism’s effect. It is not a test of whether decomposing in the first place changes upstream search or evidence-gathering behavior for the primary claim – that would require an independently-run single-claim-only pipeline configuration, which does not exist. No new LLM calls were made; restricted to the 4 items with more than one verifiable claim, since aggregation is a no-op on a single-claim item by construction.

This is a real, positive result for RQ2’s decomposition-specific question, about the aggregation mechanism specifically. On item-0001, the single highest-importance claim landed on UNVERIFIED (a missing-citation validator downgrade), but aggregating all 4 extracted claims via the deterministic rule table (Section IV-D3) correctly recovered MOSTLY_FALSE. $n = 4$ is far too small to generalize, but this is the clearest component-level evidence in this program that aggregation helps rather than hurts once claim extraction has more than one claim to work with, reported with the same prominence as Section VII’s findings, not buried beneath them.

TABLE VI
Per-item detail (✓ = correct, ✗ = wrong)

Item	Single-claim result	Multi-claim result	Ground truth
item-0001	✗ UNVERIFIED	✓ MOSTLY_FALSE	FALSE
item-0002	✓ MOSTLY_FALSE	✓ MOSTLY_FALSE	FALSE
item-0004	✗ FALSE	✗ FALSE (no-op, 1 claim)	TRUE
item-0005	✗ FALSE	✗ FALSE	MISLEADING

B. Validation ablation (RQ1) – summary

Baseline 4 (TruthLens minus validation) is the primary ablation for RQ1; its result is the unsupported-output-rate comparison in Section IX (66.7% without validation vs. 55.6% with it, on the original 9-case validator audit, before the two general fixes reported in the same section improved that further).

C. Source-tier ablation (RQ4) – summary

Restricting research-planning queries explicitly targeting primary sources to a fixed government/legal domain allow-list, versus the system’s normal unrestricted queries, is evaluated in full in Appendix A (summarized in Section IX): tier-classification rate moved from $\sim 11\%$ to 95% under restriction, measured on a different, smaller query sample than the system’s normal operating condition, with a disclosed residual gap in full-text fetch reliability specific to .gov.in domains.

IX. Evidence and Validation Analysis

A. Validator methodology

The real production pipeline was run with SKIP_VALIDATION=True so the raw proposal is always persisted, while validate_verdict()’s outcome is still always computed – giving both the WITHOUT- and WITH-validation view from one real LLM call per claim. 9 verifiable claims resolved across 4 items (0001, 0002, 0004, 0005); items 0006/0007 had zero verifiable claims (Section X). Ground truth is a single, unadjudicated draft human review – every number below is directional pending independent review. One real bug was found and fixed mid-audit: a single-bracket citation’s UUID hex fragments leaked through the number-grounding check as apparent unsupported statistics (item-0004), fixed by stripping anything UUID-shaped independent of bracket style – with a disclosed side effect (it removed an accidental protection that had been shielding a substantively wrong verdict, Section XIII).

B. Original confusion matrix ($n = 9$, single unadjudicated reviewer)

Caveat: 3 of the 4 recorded downgrades are “no-op” – the raw proposal was already UNVERIFIED with an empty citation list, so the check fires trivially without changing the published label. Counted as TN (nothing wrong was going to publish either way), not a meaningful

TABLE VII
Validator confusion matrix, before the two general fixes below

	Validator downgrades	Validator passes
Human: unsupported	1 (TP)	5 (FN)
Human: fine to publish	0 (FP)	3 (TN)

catch – some unknown share of any raw “X% downgraded” number may be this kind of no-op.

Validator Precision = $1/1 = 100\%$ (Wilson 95% CI: 20.7–100%). Validator Recall = $1/6 = 16.7\%$ (Wilson 95% CI: 3.0–56.4%). Validator F1 $\approx 28.6\%$.

C. Unsupported output rate, and why recall is low

Of the 9 raw proposals, 6/9 (66.7%) are judged, by draft human review, unsupported enough that publishing them as-is would mislead – wrong -entity evidence treated as contradicting; a label contradicting the reasoning’s own “no reliable information” conclusion; two confident labels contradicting independent Tier-1 ground truth. The original three checks catch only 1 of these 6. Unsupported output rate WITHOUT validation: 6/9 (66.7%). WITH validation: 5/9 (55.6%) – real and positive, more modest than a raw downgrade rate alone would suggest. Every one of the 5 false negatives fails for a reason citation existence, source-fetch existence, and numeric grounding structurally cannot detect: label/reasoning inconsistency (motivated the new Check 4, below); wrong-entity evidence treated as relevant; vague reasoning that still clears a word-count bar; and a verdict simply, factually wrong relative to ground truth that no citation check can catch if the model misweighs real, present evidence.

D. Addendum: re-scoring after two general fixes, with a circularity caveat

Two general, deterministic fixes were added to validate_verdict() after the headline finding in Section VII: the UUID-leak fix above, and the new label/reasoning-consistency check (Section IV-D1) – both pure functions of the model’s own output, never of ground truth, so re-scoring the same 9 cases costs no new LLM calls and is not answer-tuning.

Precision 100% (2/2), Recall 40% (2/5) – up from 16.7%. The two new true positives are the two most consequential original false negatives, including a confident FALSE at confidence 0.8 that contradicted independent Tier-1 ground truth, now correctly downgraded. A real

TABLE VIII
Validator confusion matrix, after the two general fixes

	Validator downgrades	Validator passes
Human: unsupported	2 (TP)	3 (FN)
Human: fine to publish	0 (FP)	4 (TN, all no-op)

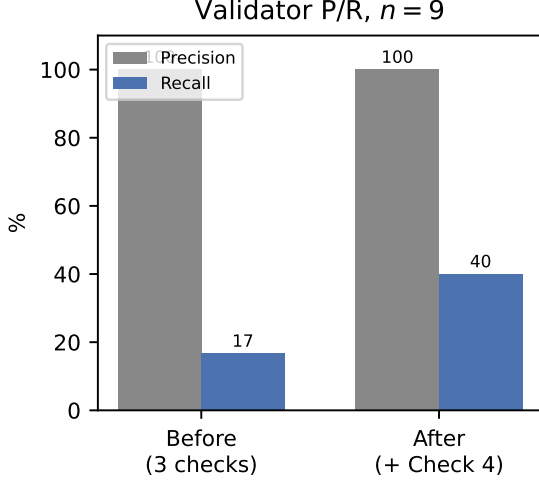


Fig. 5. Validator precision/recall before vs. after the two general fixes (Tables VII and VIII, $n = 9$). Precision holds at 100%; recall improves 16.7%→40% – with the calibration circularity disclosed in the text.

regression, reported with equal prominence: the audit’s one original true positive is no longer downgraded, since the UUID-leak fix removed the false trigger that had been accidentally catching it – the underlying wrong-entity problem remains genuinely uncaught; net effect 1 TP lost, 2 gained, landing at 2 TP / 5 FN.

A disclosed circularity, now tested rather than left as a suspicion: the new check’s phrase list was written after reading the two cases it now catches, so 40% is likely optimistic about generalization. Four real paraphrases never in the original phrase list were each tested against a confident label in a synthetic evidence matrix (VALIDATOR_SYNTHETIC_BENCHMARK.md, full phrasings in Appendix A); all four were missed – Check 4 does not generalize beyond its literal phrase list, confirming the threat to validity already disclosed (Section XIV).

Reel-level accuracy did not change (2/6) despite this real validator improvement – itself a finding, not a disappointment to explain away (Section XV).

E. Evidence quality (RQ4): tier, stance, and diversity distribution ($n = 68$ sources, 9 claims)

Source-tier classification rate (primary tiers combined): $16/68 = 23.5\%$ (Wilson 95% CI: 15.0–34.9%) on this sample’s normal, unrestricted queries – a different experimental condition from the domain-restricted result below, not a contradiction.

TABLE IX
Source tier distribution ($n = 68$)

Tier	Count	%
other	43	63.2
primary_government	14	20.6
established_news	6	8.8
primary_legal	2	2.9
news_wire	2	2.9
factcheck_org	1	1.5
primary_data	0	0.0

TABLE X
Evidence stance distribution ($n = 68$)

Stance	Count	%
irrelevant	58	85.3
contradicts	8	11.8
supports	2	2.9
provides_context	0	0.0

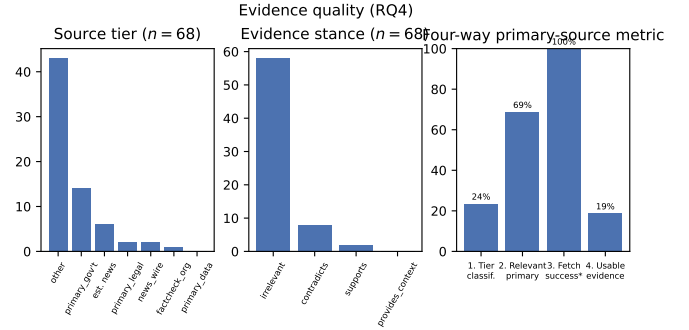


Fig. 6. Source-tier distribution, evidence-stance distribution, and the four-way primary-source metric ($n = 68$ sources, 9 claims). *Metric 3 holds by construction of an existing fetch guarantee, not independently re-measured this pass.

85.3% of retrieved-and-analyzed sources were judged irrelevant – notably higher than an earlier, smaller development-time figure of 55.6%, both real and reported since this ratio moves across samples. Source diversity: 29 distinct domains across 68 sources.

F. The four-way primary-source metric, not collapsed into one number

Four distinct, independently reported metrics, not one collapsed number (full per-metric breakdown and illustrative wrong-domain/wrong-country examples: Appendix A): (1) source-tier classification rate $16/68 = 23.5\%$; (2) relevant primary source rate (draft human judgment) $11/16 = 68.75\%$; (3) primary source fetch-success rate $16/16$, by construction of an existing fetch guarantee; (4) usable evidence extraction rate $3/16 = 18.75\%$ (stance $\neq$ irrelevant).

The gap between metric 2 (68.75%) and metric 4 (18.75%) is the single most important finding of this evaluation. Retrieval finds real, topically-appropriate primary sources at a meaningfully higher rate than the

usable-evidence number would suggest – the bottleneck is that most of these sources are homepages or generically-relevant documents too broad to confirm one specific atomic fact, not a failure to find relevant sources. Metric 4 alone would read as poor retrieval quality; both metrics together show the more accurate picture.

G. A second major bug, a search-side fix, and an unintegrated entity-consistency prototype

Three further findings from this same evidence-quality audit are reported in full in Appendix A, summarized here: (1) 32/68 (47.1%) real evidence rows had an empty explanation field despite a valid stance label – the same schema-valid-but-empty class found elsewhere, never guarded at this stage, fixed with a per-source substantiveness check (Section IV-D2); (2) restricting primary-source-targeting queries to a fixed government/legal domain allow-list, rather than trusting a model-written site:gov.in, moved tier-classification from $\sim 11\%$ to 95% on seven re-run queries, with a disclosed residual .gov.in full-page-fetch reliability gap; (3) a prototype entity-consistency check (does a claim’s extracted entity appear in its cited evidence?) found one genuine true positive on the real 9-claim sample – refining item-0004’s error attribution to a concrete evidence mismatch (Burdwan, West Bengal vs. Delhi Police) – alongside diagnosed false-positive causes, and is explicitly not integrated into production: too small a sample to justify a fifth check without more data (Rule 4).

X. Multimodal and Visual Failure Analysis

A. Defining claim coverage

Every use of “claim coverage” means one operationalized judgment: a benchmark claim is covered when at least one extracted claim identifies the same factual proposition as the professional fact-check’s own claim – matching on subject/entity, event, predicate, and any temporally-relevant qualifier – judged by a human annotator blind to which pipeline configuration produced the extraction (METRICS.md §2). Topical overlap alone does not count. This is necessarily a human judgment, not a string-similarity score, since paraphrase makes automated matching unreliable at this scale; every pairwise judgment is recorded in research/claim_coverage_results.csv for audit.

B. What was run

Real ingestion of the 7-item dataset as it existed at the time, followed by claim extraction under three input-modality conditions: text_only, text_ocr (+ on-screen text), text_ocr_vision (+ vision-context description) – all local-model-only, isolating the modality variable the same way the baselines isolate architecture. TruthLens has exactly three input signals feeding claim extraction, so text_ocr_vision is full multimodal for this system. Ingestion: 6 of 7 items usable – item-0003 failed both attempts

TABLE XI
Claim coverage by input modality

Condition	Covered	Coverage	95% CI
text_only	1/6	16.7%	[3.0,56.4]
text_ocr	0/6	0.0%	[0.0,39.0]
text_ocr_vision (full)	2/6	33.3%	[9.7,70.0]

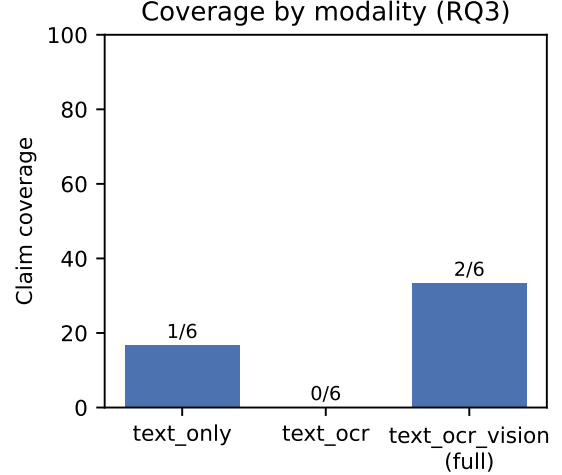


Fig. 7. Table XI plotted. text_ocr scoring below text_only is reported as a real result, not suppressed (Section X explains the likely cause).

(persistent Instagram-side error); item-0005 initially failed on a differently-phrased yt-dlp “no video” message, fixed and re-verified live (Section XIII).

C. Claim coverage results ($n = 6$, item-0003 excluded)

Read these as exactly what they are: 6 single, non-repeated LLM calls per condition, CIs wide enough to overlap almost entirely. text_ocr scoring below text_only is real, reportable, not suppressed for complicating the “more modality is better” hypothesis – the likely (unverified) explanation is single-call noise compounded by a specific extraction failure, not a robust modality effect.

D. item-0004: the one clean, positive RQ3 result

The real transcript is badly garbled (chaotic protest audio); real OCR, sampled across frames, clearly and repeatedly captured on-screen text asserting the actual check-worthy claim (police using a nail-fitted baton), matching the fact-check’s own verdict almost verbatim. text_only extracted nothing resembling it; text_ocr happened to return one empty-text claim on this run; text_ocr_vision correctly surfaced the claim, messily (7 near-duplicate claims, one per noisy OCR-frame variant – a new, unfixed failure mode, no existing check evaluates cross-claim redundancy). Only the multimodal condition produced any usable signal, via OCR content the audio transcript had no access to – a genuine, small- n demonstration of what RQ3 asks about.

E. The cross-post attribution problem

Drafting atomic claims against the real ingested content surfaced a structural pattern named explicitly here: the cross-post attribution problem. The same underlying media M can be published as Post A asserting claim C , and separately as Post B asserting a different claim C' – a system ingesting only one post URL cannot discover the other claim from that post’s own content at all. 4 of 6 scoreable items have their actual false claim living entirely outside the specific post’s own content, added by a different account recirculating the same video under a new caption; only item-0004 (and partially item-0007) have the checkworthy claim actually on-screen in the URL this dataset points to – structural, not a sampling accident, and TruthLens has no reverse-video/cross-post matching capability. Full formalization and discussion of why this is structural rather than incidental, and why we did not prototype a fix against this program’s own benchmark: Appendix A.

F. New failure modes found this pass

Near-duplicate claims from noisy OCR-frame variants (item-0004, above), uncaught since no existing check evaluates cross-claim redundancy. importance values outside its declared $[0,1]$ range causing schema-validation failure (item-0005: -1 , then a run of 2–6 resembling an ordinal position substituted for a score) – occurred only because this script removes the Gemini -escalation cascade to isolate the modality variable, so production’s Fallback-LLMProvider would likely mask this normally. The vision-output prompt-echo bug recurred 4 more times (items 0002, 0004, 0006; items 0001, 0007 produced empty vision output instead) – a repeated, frequent vision-stage failure, not a rare anecdote, motivating the general fix in Section IV-D2. Only item-0005’s vision output was genuinely coherent. This fix’s real-world impact on Section VII’s numbers is not independently re-verified end-to-end (the paid fallback’s quota was already exhausted before it could be exercised against items 0006/0007, whose underlying problems are not primarily vision-related, so no change is expected – an expectation, not a verified result).

XI. Bias, Calibration, and Efficiency Analysis

A. Political bias (RQ5): attempted, blocked by data availability, deferred honestly

No genuine matched pairs exist in the current 9-item dataset (same topic, structure, evidence availability, and difficulty, varying only the political actor named): BJP has the most items (3) of any actor, but none are topic-matched against another single actor – a disclosed methodological choice trading this analysis away for broader dataset coverage, not attempted as a quantitative result. Named as the top open item for whoever continues this program (Section XVI). One qualitative data point (not a substitute for the real analysis, full discussion: Appendix A): the system was willing to reach FALSE

TABLE XII
Efficiency, real measured data

Config	LLM calls	Latency	Cost
B2: Search+LLM	1	~15.9s	\$0
B3: Search+RAG+LLM	1	~17.8s	\$0
Full TruthLens	~9.6/claim [‡]	n/m	\$0/paid mix

[‡]Derived from 7.56 sources/claim (avg.), each requiring one evidence-analysis call, plus one research-planning and one verdict call per claim. n/m = not measured in this pass (disclosed gap, below).

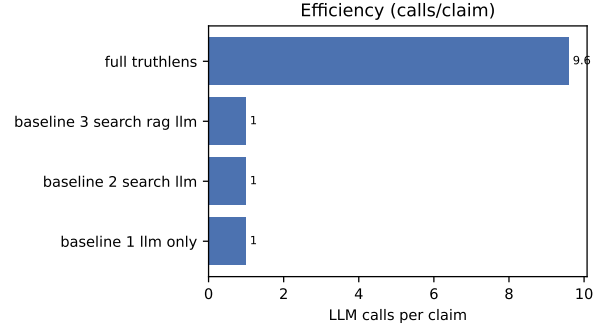


Fig. 8. LLM calls per claim by configuration (research/system_efficiency.csv). Full TruthLens’s call count is derived from real average sources/claim, not directly measured end-to-end; latency is not shown, a disclosed gap (text).

against a claim traceable to a government actor’s own public position (item-0004), a necessary but nowhere near sufficient condition for what RQ5 actually asks.

B. Calibration (RQ6): not computed, per our own pre-registered fallback rule

Nine real confidence values exist: 0.0, 0.0, 0.1, 0.1, 0.2, 0.2, 0.2, 0.7, 0.8 – too few per bin to meet our pre-registered calibration gate (no bin under 3 items) no matter how bins are drawn, so per our protocol’s fallback rule, no ECE or Brier score is reported. One heavily caveated observation on this raw distribution: Appendix A.

C. Efficiency

Full TruthLens is not directly latency-comparable to the two baselines – a disclosed gap, since the validator-audit run did not capture per-claim wall-clock time. What is measurable: full TruthLens’s real cost/time per claim is necessarily several times either baseline’s from call count alone – roughly 9.6 LLM calls per claim against the baselines’ fixed 1. Full latency/cost gap detail: Appendix A.

XII. Foundation-Phase System Extensions

A subsequent work program, run against a formal 13-phase research roadmap (research/RESEARCH_ROADMAP_V2.md), extended the dataset and system described above. Scope discipline, stated up front: none of this work re-measures reel-level

accuracy or touches the frozen paired six-item comparison in Section VII – the new benchmark items below are held in a validation split, explicitly not used for any tuning decision, and the formal DEV/VALIDATION/TEST freeze (this roadmap’s own Phase 12) has not yet occurred. What follows is real, measured, component-level work, reported with the same discipline as everything above: every number traces to a versioned experiment in `experiments/registry.jsonl` (EXP-009 through EXP-029), and negative or inconclusive results are kept exactly as prominent as positive ones.

A. Dataset growth: 9 → 15 items

Six new items were sourced from the same professional fact-checking pool (BOOM Live, Alt News), each independently verified: a live Instagram URL check, and content-hash deduplication against every existing item. Composition shifted from 7 FALSE / 1 MISLEADING / 1 TRUE to 12 FALSE / 2 MISLEADING / 1 TRUE (still no second TRUE-labeled item despite two dedicated sourcing passes – consistent with, not a failure of, the already-disclosed structural skew in Section VI-E: professional fact-checkers publish far more debunks than confirmations). All six new items were fully ingested (real transcription, OCR, vision-context) and assigned to a validation split held separate from the original nine’s dev split – the first genuine split discipline this project has had, short of the ~20–30 item target Future Work already named.

B. Two new deterministic validator checks, integrated with measured evidence

Two new checks were added to `validate_verdict()`, each following the same discipline as the original four (Section IX): prototyped independently, not integrated until real evaluation cleared a stopping-condition bar.

Check 6 (temporal consistency) flags a verdict when the claim asserts an explicit, unambiguous date and every cited source with a known publication date pre-dates it by more than two days – the “old footage presented as current” pattern. Deliberately narrow: real `time_reference` values in this project’s data are almost all vague (“recent,” “present,” “unspecified”); only explicit dates resolve, avoiding false positives on legitimately time-unanchored claims.

Check 7 (entity consistency) flags a verdict when none of the claim’s PERSON/ORGANIZATION/LOCATION entities appear, exactly or via a calibrated fuzzy match (`difflib.SequenceMatcher`, threshold 0.75, tuned against real transliteration-variant pairs in this project’s own data), in a cited source with a stance that actually influenced the verdict. This is a direct promotion of the entity-consistency prototype named as unintegrated future work in an earlier pass of this program: the original prototype’s 4 false positives (all one abstract-concept entity, {“Democracy”, type: “concept”}, matched against unrelated legal text) motivated a type filter; a corrected

re-evaluation against real evidence found 2 of 2 flagged violations were genuine true positives with zero false positives, clearing the integration bar.

Measured effect: the adversarial synthetic validator benchmark (Section IX) was expanded from 28 to 34 cases with six new cases targeting exactly these two checks, including a direct re-test of the original benchmark’s own previously-diagnosed, explicitly-marked-uncatchable entity-mismatch case. Precision rose 81.8%→87.5%, recall rose 50.0%→60.9%, and the previously-named gap case is now caught. The benchmark’s only two false positives are the same, already-documented Check-3 limitation from the original 28 cases (unrelated to Checks 6/7) – both new checks add real recall with zero measured precision cost.

C. A cross-cutting reliability finding, confirmed four independent times

Four separate experiments, each targeting a different pipeline stage, independently surfaced the same underlying pattern: raw, un-retried output from the local model (llama3.2) on real content is substantively empty or unusable a large fraction of the time, and this system’s existing layered defenses (Gemini-escalation quality retries) are doing substantially more real work than a pipeline-level pass/fail reading would suggest.

- Claim source_quote population: 0 of 12 real extracted claims across 8 real items had a verbatim-groundable quote (EXP-012); independently reconfirmed at 0 of 90 claims across a full 8-way modality sweep (EXP-016, Section XII-D).
- Evidence-stance explanations: 80% (16/20) of raw evidence-stance judgments, run without the existing substantiveness-retry safeguard, had an empty explanation on at least one side of a taxonomy comparison (EXP-017).
- Adversarial stress testing (EXP-020, Section XII-D): zero crashes across six adversarial text-content cases, but the one case needing a real quality-retry hit real Gemini quota exhaustion from this same evaluation program’s own cumulative usage.

No single one of these experiments was designed to measure model reliability directly; each surfaced it as a byproduct of testing something else, which is why four independent confirmations are more convincing than one dedicated study would be. Full detail: Appendix A.

D. Retrieval, multimodal, and cross-post extensions (summary)

Three further extensions were built and measured, each with a real, disclosed result rather than an assumed win (full numbers: Appendix A):

Structured 5-query retrieval (exact-claim, entity-focused, primary-source, contradiction, context-history, replacing the prior loosely-typed 2–5 query generation) raised source-tier classification (23.5%→38.1%, $n = 42$ sources / 4 claims) but lowered manually-judged

relevance (68.75%→43.75%) and usable-evidence rate (18.75%→0%) on this small sample – traced to a real, disclosed confound: one claim with no natural institutional subject (a private organization) received 7 of 16 primary-tier hits, all irrelevant, because the mandatory primary-source query type fires unconditionally. Excluding that one claim, the other three score 78% relevant, above baseline. Reported as a genuine mixed result, not smoothed into a clean win.

Full 8-combination multimodal re-analysis (extending Section X’s 3-condition design to independently toggle audio/OCR/vision, caption always on) reproduced rather than resolved the original non-monotonic surprise: vision-alone scored highest on ground-truth claim coverage (37.5%), and ocr+vision scored below every single-signal condition (12.5%), on the same 8 real items.

Cross-post detection, stage 1 of 3: a perceptual-hashing first-pass filter (imagehash, a new dependency) was built and validated against real re-extracted video frames – a same-video control matched at Hamming distance 0, a different-video control did not match (distance 20, threshold 10). Stages 2 (claim-embedding clustering) and 3 (Gemini escalation for ambiguous pairs) are not yet built.

Bounded adversarial stress testing of claim extraction (six hand-constructed cases: garbled OCR, caption/transcript contradiction, near-empty input, mixed-script chaos, repetitive content, near-max-token length) produced zero crashes and no hallucinated cross-signal merging, but surfaced two previously-unnamed structural gaps: no check anywhere tests whether a post’s caption and transcript are mutually consistent, and mixed-script content showed a possible bias toward extracting only Latin-script segments, silently dropping non-Latin-script content from the same continuous statement.

E. Immediate follow-up: two named gaps investigated, one closed negatively, one corrected

Both gaps named above were investigated in the same autonomous pass, each with a real, disclosed outcome rather than left standing as untested hypotheses (full detail: Appendix A).

The input-signal consistency check was found infeasible in its obvious cheap form: keyword-overlap (Jaccard) similarity between a post’s caption and transcript, calibrated against real data before being written into production, is 0.000 for every real, legitimate benchmark item checked ($n = 4$) – captions and transcripts are different text registers, and often different languages, even when fully consistent. A check calibrated this way on this project’s own data would flag essentially all legitimate content as inconsistent. Real semantic understanding would be needed instead of a cheap heuristic.

The multilingual-extraction bias suspected above does not generalize: re-examined against the larger 90-claim dataset already collected for the 8-combination multimodal analysis (Section XII-D), non-Latin-script claims

are the majority (48/90, 53.3%), and script distribution tracks each real item’s own dominant language exactly – including one genuinely code-switched real item, which produced claims in both scripts, not just Latin. The original single-case finding was real but did not generalize; the likely correct, narrower reading is that unnaturally dense intra-sentence code-switching (three scripts within one synthetic clause) is harder than the inter-sentence or paragraph-level code-switching that occurs in real speech, not a general script bias. We report this correction with the same prominence as the original finding, per this paper’s standing discipline.

Separately, the source_quote population gap (Section XII-C) was directly attacked with a strengthened extraction prompt – an explicit self-check procedure plus a worked example built from a real observed miss (a claim whose text was itself a verbatim OCR line, “5,154 likes”, yet left unquoted). The fix did not work: 0% source-quote population both before and after, $n = 8$ items, and on the exact item the worked example was built from, the strengthened prompt produced zero claims where the baseline had produced seven unquoted ones. This is a clean negative result for prompt engineering as the fix, strengthening rather than resolving the underlying finding.

Finally, the 6 real VALIDATION-split items (Section XII) were, for the first time, run through the complete real pipeline (claim extraction through verdict) and the results genuinely committed – unlocking the aggregation counterfactual this paper’s Future Work has named since Section VIII-A. 4 of 6 items produced zero claims (consistent with this pass’s own already-documented low real-world extraction yield); the other two produced 3 real claims with real verdicts. One of them is independently notable: Check 7 (entity consistency) correctly downgraded a claim about a Jharkhand-police protest that had been cited against a wholly unrelated real article (a 2026 IIT Palakkad security incident) – the first live confirmation that Check 7 generalizes to genuinely new production data, not just the historical and synthetic cases it was built and tested against.

F. A second autonomous pass: aggregation confirmed, a security defense measured and hardened, a verdict-stage reliability gap found

A further autonomous pass extended this work along four lines, each producing a real, disclosed result rather than an assumed one.

Aggregation (Phase 10), previously blocked on the VALIDATION split having no multi-claim items with real verdicts, was unblocked by the population pass above (Section XII-E): exactly one VALIDATION item (item-0010) qualified. Single-claim-only resolved to UNVERIFIED (a miss against real ground truth FALSE); full decomposition, using the same `derive_overall_verdict()` production itself calls, correctly resolved to FALSE. Directionally consistent with the original $n = 4$ DEV-split finding

(Section VIII-A) that aggregation helps, but explicitly reported at its real $n = 1$ and not claimed as statistically meaningful.

Cross-post detection stage 2 (claim-embedding clustering, Section XII) was attempted and found blocked by a concrete, confirmed infrastructure gap rather than left unattempted without explanation: this project’s local Ollama server was not started with embedding support (`ollama.Client().embeddings(...)` fails with an explicit “start it with `-embeddings`” error, confirmed live), and no dedicated embedding model is pulled locally either. Restarting a service this project depends on, or pulling a new model, was judged outside what an autonomous pass should do unilaterally – disclosed as a real, concrete prerequisite rather than worked around with a weaker substitute under the same column’s name.

Prompt-injection defense had zero existing test coverage despite processing genuinely untrusted content (transcript, OCR, caption) by design on every real post – confirmed by an empty result from searching the test suite for any injection-related coverage before this pass. Two things were found and both were fixed, not just reported. First, code review (not model behavior) found the delimiter convention every LLM stage relies on never neutralized literal occurrences of its own delimiter tokens already present in untrusted text – a deterministic, model-independent bypass, closed by sanitizing those tokens before wrapping and verified in both directions. Second, 5 live adversarial attempts against claim extraction found the most direct injection style (a blunt override instruction) succeeded once, producing the injected false claim verbatim as a verifiable, maximum-importance claim – the first real measurement of this project’s own previously-stated expectation that small local models resist in-band instructions less reliably than larger ones. A hardened extraction prompt reduced this to 0/5 on re-test. Separately, carrying the one pre-fix successful injection through the real remainder of the pipeline (research planning through verdict) showed the downstream anti-hallucination validator independently resolving it to a correct, evidence-grounded FALSE verdict citing 12 real sources – confirming, for the first time with a measurement rather than an assertion, that this system’s documented security posture (the validator as the actual backstop) holds for at least this case. Full detail, including the caveat that this is $n = 1$ against an unusually well-evidenced test claim: `research/PROMPT_INJECTION_STRESS_V2.md`.

A new adversarial category (Step 18’s #8, “contradictory sources”) at the verdict stage surfaced a real, reproducible failure not previously characterized: constructed evidence with a 0.95-reliability primary-government source supporting a claim and a 0.20-reliability, uncited source contradicting it produced TRUE or MOSTLY_TRUE in 0 of 14 real trials across two prompt versions – including runs whose own reasoning text correctly identified the contradicting source as unreliable while still not choosing

a label reflecting that assessment, and one run pairing a wrong label with maximum stated confidence. An explicit reliability-weighting rule added to the verdict prompt, with a worked example distinct from the held-out test case, did not resolve this on re-test (still 0/7). This is reported as a genuine, measured negative result – the fix is kept regardless, since it is objectively correct guidance and not shown to make anything worse, but is not claimed to have solved the underlying issue. The likely real fix, a deterministic validator check comparing cited -evidence reliability against verdict direction (mirroring how Checks 6 and 7 succeeded where prompt-only approaches would not have been trustworthy), is named as a concrete candidate for a dedicated future pass rather than added without its own integration evaluation. Full detail: `research/CONTRADICTORY_SOURCES_V2.md`.

Finally, a direct re-run of the evidence-stance taxonomy comparison (Section XII-E named this as the concrete next step) fixed its own named confound – porting the production substantiveness-retry safeguard into the comparison’s new-taxonomy call – and in doing so surfaced a deeper, quantified constraint rather than a clean answer: a real HTTP 429 response confirmed the fallback model’s free tier is capped at exactly 20 requests/day on this project’s key, a limit this session’s own real usage repeatedly exhausts. Three independent attempts (4/20, 6/20, 5/20 clean rows) remain inconclusive about whether the candidate taxonomy adds real value – not from an uncontrolled confound anymore, but from a hard, external infrastructure ceiling, now named and quantified rather than merely suspected.

XIII. A Taxonomy of Failure Modes Found in Live Testing

Running the full system against real content end-to-end – both during original development and during this evaluation program itself – surfaced 24 specific, reproducible failure modes we believe are under-documented in the automated fact-checking literature at this level of implementation detail, because they only appear when a system is run against real content rather than evaluated on a static benchmark. We consider this taxonomy, and the general lesson that most of these failures were discoverable only this way, to be as practically important a contribution of this work as any single architectural component. The full list, each entry with its fix, is given in Appendix A; six representative entries, spanning production code, research tooling, security, and this program’s own benchmark construction:

- Schema-valid, substantively empty output, found independently in four different pipeline stages (claim extraction, content generation, verdict reasoning, evidence analysis) – no schema validation catches it; only a content-level substantiveness check does (Section IV-D2), and each stage needed its own.

- Entity confusion between similarly-named organizations and the downstream failure it caused: a downgraded verdict’s hallucinated free-text reasoning was still reused verbatim on a rendered slide, because downgrading forces only the label, not the text.
- Baseline architecture silently inheriting a rescue mechanism under evaluation (Section V) – an early draft of Baselines 2–3 called the same provider factory every production stage uses, giving both baselines the exact failure-rescue mechanism TruthLens’s own architecture was being evaluated for having.
- A table-generation script’s printed output and its persisted JSON summary silently disagreed, found while building this paper’s own figures (day8_final_tables.py reused two variable names across three unrelated computations) – the same failure class as the production-code entity-confusion entry above, found in this project’s own research tooling instead.
- Six synthetic validator benchmark cases silently tested the wrong check (Section IV-D1/Phase 4, VALIDATOR_SYNTHETIC_BENCHMARK.md) – cases built to isolate Check 4 had no citation at all, so Check 1 fired first and Check 4 was never reached, caught only by reading validation_status in the raw output rather than trusting the benchmark’s own design.
- A blunt prompt-injection override got past claim extraction once in 5 live attempts, producing the injected false claim verbatim as a verifiable, maximum-importance claim – the first real measurement (not just a stated expectation) that this system’s delimiter-based defense is imperfect against small local models, confirmed independently by the downstream validator resolving the same injected claim to a correct FALSE verdict when carried through the rest of the real pipeline.
- The vision-context fix’s effect on headline accuracy is not independently re-verified – stated as an expectation, not a measured result.
- item-0003 is permanently excluded, not scored as a failure – confirmed unfetchable across three attempts.
- Baseline comparison isolates architecture from model, but not from claim-extraction coverage – Baselines 1–3 receive TruthLens’s own already-extracted claim, so the headline comparison cannot fully separate architecture underperformance from claim-extraction giving it nothing to work with.
- The primary-source-retrieval fix is measured on tier classification, not full-text completeness, on a distinct query sample; the .gov.in fetch-reliability gap it surfaced remains open.
- RQ5 and RQ6 are deferred by design, not omission, per pre-registered fallback rules (Section XI).
- Ground truth is sourced from a small set of professional fact-checkers whose editorial judgment we do not independently audit – a systematic coverage or resolution asymmetry would propagate undetected, since no second ground-truth source exists to check against.

XV. Discussion

On the value of auditing your own baselines as skeptically as your own system. This program’s most consequential finding was produced not by improving TruthLens but by a post-hoc audit that treated our own experimental design with the same skepticism this paper applies to the system under test, finding that our baselines had been quietly advantaged for the entire program (Section VII-A). Without that audit, this paper would have published a confident, wrong -in-degree explanation for why TruthLens loses to a single-shot baseline: the diagnosed causes (claim-extraction coverage, validator false negatives) were real contributing factors, but the uncorrected number overstated their effect, since part of it was an artifact of unequal claim input, not architecture. The general lesson we think generalizes beyond this paper: a baseline is part of the experimental apparatus, not a neutral reference point, and deserves the same implementation-level scrutiny as the system it is compared against.

On label accuracy vs. support validity. Two general, ground-truth-independent fixes moved validator recall from 16.7% to 40% (Section IX) while reel-level accuracy stayed at 2/6 throughout, both before and after the baseline correction. Label accuracy and support validity are different axes that can move independently – a concrete demonstration, not an inference from theory. A benchmark reporting only label accuracy would have shown no progress during a pass of work that materially reduced how often the system publishes a confidently wrong, unsupported verdict.

The corrected comparison, and when TruthLens helps specifically. Once claim input is held constant, full Truth-

XIV. Threats to Validity

Ten threats, stated here at the density needed to be checkable; full reasoning for each: Appendix A.

- Small, non-independent sample. $n = 9$ (6 resolved from every configuration), below this program’s own 20–30 target; several proportions, including the headline 33.3% accuracy figure, have Wilson intervals wide enough to be consistent with a different true rate.
- Single domain, single-country content. Indian political short-form video only; source-tiering and prompts untested elsewhere.
- Single annotator, no inter-annotator agreement on every draft judgment underlying Sections IX/ X (Tier-1 headline verdicts are exempt – they are independent organizations’ own published verdicts).
- The new validator check’s 40% recall figure is measured on the same sample it was calibrated against – likely optimistic about generalization (Section IX).

TABLE XIII
Per-item conditions, corrected comparison ($n = 6$)

Item	Multi-claim?	Claim extracted?	TruthLens vs. B2
item-0001	Yes (4)	Yes	Both correct
item-0002	Yes (2)	Yes	Both correct
item-0004	No (1)	Yes	B2 only correct
item-0005	Yes (2)	Yes	Both wrong
item-0006	No (0)	No	Both wrong (TruthLens abstains)
item-0007	No (0)	No	Both wrong (TruthLens abstains)

Lens beats two of three baselines outright and trails the third by a much smaller margin than before (Section VII); the two remaining wrong items trace to claim-extraction coverage failure (items 0006/0007) and one validator false negative (item-0004), and Section VIII-A shows aggregation delivering a real gain once claim extraction has more than one claim. Table XIII replaces a single win/lose framing with the most direct answer this sample can give to where the advantage materializes.

Reading the matrix rather than the aggregate: TruthLens and Baseline 2 agree on every item where claim extraction succeeded and produced more than one claim (items 0001, 0002) – exactly where Section VIII-A’s counterfactual shows aggregation adding value. TruthLens’s one exclusive loss (item-0004) is a single-claim item, where decomposition has nothing to contribute and the comparison reduces to per-claim reasoning quality alone, where the already-diagnosed validator false negative costs it. Both systems fail identically whenever claim extraction itself fails (items 0006/0007) – architecture downstream of claim extraction cannot compensate for having no claim, for either system. The pattern this small sample suggests, stated as a hypothesis for a larger dataset to test, not a conclusion this n can support: TruthLens’s advantage over a single-shot baseline may be concentrated specifically in multi-claim items with successful extraction, and structurally absent everywhere claim extraction fails or a post has only one claim to begin with.

This also changes how DeVerna et al.’s finding [10] that curated context matters more than reasoning ability for political fact-checking should be read against our results. The corrected comparison is broadly consistent with theirs: TruthLens’s curation-heavy architecture is competitive with, and mostly ahead of, simpler alternatives once claim input is fairly controlled, and its one remaining loss traces to a specific, upstream coverage gap rather than curation itself underperforming. The lesson we draw is methodological as much as architectural: curation-heavy systems should be evaluated at the component level, with baseline implementation audited as carefully as the system under test, because an end-to-end label-accuracy number alone – as this program’s own uncorrected first pass demonstrated – can misattribute a claim-input confound to an architectural one.

Finally, we note what this program deliberately did not do: it did not tune any prompt, threshold, or model choice

on the frozen held-out dataset’s contents (Section V), and the two general validator fixes were built to be ground-truth independent, with the one place that assumption is imperfect (the new validator check’s phrase-list calibration) disclosed as a named threat to validity rather than presented as a clean win. The baseline correction in Section VII-A is similarly ground-truth-independent: it changes what claim text baselines receive, not any threshold or judgment tuned against knowing the right answer. We would rather report a real 33.3%, a real correction, and an honest account of both than a smoother-looking number we could not defend the provenance of.

XVI. Future Work

- 1) Scale the held-out benchmark past $n = 9$ – partially addressed: grew to 15 (Section XII), still short of the 20–30 target and the ~ 40 –60 needed for real McNemar power; the $\sim 8\%$ Tier-1 sourcing hit rate remains the binding constraint on how fast this can scale further.
- 2) Diagnose and close claim-extraction coverage failures on degraded audio. Still open – EXP-016 (Section XII-D) re-confirmed rather than closed this gap: the same items still produce zero verifiable claims across all 8 modality combinations tested, not just the original 3.
- 3) Video-provenance verification. Partially addressed, stage 2 blocker now diagnosed: a perceptual-hashing first-pass filter is built and validated against real video frames (Section XII); stage 2 (claim-embedding clustering) was attempted and found blocked by a confirmed infrastructure gap, not an unattempted item – the local Ollama server lacks embedding support and no embedding model is pulled locally (Section XII-F). Restarting the service or pulling a new model is the concrete, named prerequisite; stage 3 (Gemini escalation) and any live pipeline integration remain undone.
- 4) Test the new validator checks against phrasings/scenarios they were not calibrated on – addressed for the two checks added this pass (Section XII-B); the original four checks’ own circularity (Section IX) is still open.
- 5) Complete items 0008/0009’s full-TruthLens condition and re-verify the vision-context fix end-to-end, both blocked by the same Gemini free-tier quota exhaustion, once quota resets.
- 6) A dedicated matched-pair sourcing pass for RQ5, sourced for topic/structure/difficulty match across political actors rather than actor diversity (Section XI).
- 7) Calibration, once the dataset is large enough to bin – Section XI’s pre-registered gate (no bin under 3 items) was not met at $n = 9$ and remains unmet at $n = 15$.

- 8) A second, independent annotator, for a real inter-annotator agreement statistic on Tier-2 items and the draft judgments underlying Sections IX and X.
- 9) The .gov.in full-page-fetch reliability gap (403s, TLS failures) remains open; closing it needs a scoped certificate-verification exception or further diagnosis.
- 10) Near-duplicate OCR-frame-variant claim deduplication (Section X) – addressed: a deterministic, similarity-calibrated deduplication pass now runs before persistence.
- 11) Diagnose why claim extraction almost never populates a verbatim source_quote (Section XII-C) – attempted: a strengthened instruction (an explicit self-check procedure plus a worked example built from a real observed miss) did not move the population rate at all (0% both ways, $n = 8$ items) – a clean negative result, not an unattempted item anymore. The two live hypotheses (prompt under-elicitation vs. genuinely unquotable paraphrases) remain undistinguished; a bigger architectural change (e.g. eliciting the quote before the claim text) is the next candidate, not attempted.
- 12) An input-signal consistency check for claim extraction (does a post’s caption agree with its transcript/OCR content) – investigated: the obvious cheap approach (keyword-overlap similarity) is not viable, calibrated against real data before being written into production – Jaccard similarity between caption and transcript keywords is 0.000 for every real, legitimate benchmark item checked, because captions and transcripts are different text registers (and often different languages) even when fully consistent. Real semantic understanding would be needed instead, not attempted.
- 13) A multilingual-extraction audit (does non-Latin-script content get dropped disproportionately) – investigated and corrected: the single adversarial case motivating this item (Section XII-D) does NOT generalize when checked against the larger 90-claim dataset already collected this session – non-Latin-script claims are the majority (48/90), tracking each item’s own real dominant language correctly, including in a genuinely code-switched real item. The suspected bias was a real finding about ONE synthetic, intra-sentence code-switching construction, not a general property of this system – corrected here rather than left standing as originally (over-)stated.
- 14) Complete the full end-to-end adversarial suite (Step 18’s 20 categories, 100+ cases) – extended: beyond the original 6-case claim-extraction-only slice, prompt-injection robustness (5 cases, claim-extraction and verdict stages) and contradictory sources (#8, 3 cases plus repeated-sampling stability runs, verdict stage) are now covered (Section XII-F); categories needing real AI-generated-image/edited-

video/audio-corruption content remain unbuilt.

- 15) A deterministic reliability-weighted verdict check. A measured, reproducible failure (Section XII-F): verdict-label selection does not reliably favor a high-reliability source over a low-reliability, uncited one under direct contradiction, and a targeted prompt fix did not resolve it (0/14 real trials correct, before and after). The likely real fix – comparing cited-evidence reliability against verdict direction as an 8th deterministic validator check, mirroring Checks 6/7’s successful pattern – is named but not built, pending its own integration evaluation.
- 16) A properly isolated test database for Gemini-quota scenario tests, or a way to scope the provider’s global cooldown check to a test context – the collision between real concurrent usage and mocked scenarios (Appendix A, entry 22) is now quantified (a confirmed 20-requests/day free-tier cap) but not fixed.

XVII. Conclusion

This paper reports the first evaluation of TruthLens against independent, held-out ground truth, and we lead with the least flattering result it produced – not a weakness in TruthLens, but in our own experimental design: a skeptical post-hoc audit found that our baselines had been given a cleaner claim input than TruthLens’s own extraction ever produced, across this entire program, contradicting our stated methodology and inflating baseline performance. We report this first because a research program that only audits the system under test, never its own baselines, is not applying its own standard uniformly. Corrected, full TruthLens beats two of three baselines outright and trails the third by a substantially smaller margin than the uncorrected comparison showed, on the same paired 6-item sample; the remaining gap traces to claim-extraction coverage failures on 2 items and one validator false negative, both independently diagnosed – more scientifically useful than either the original, partly-artifactual negative number or an uncritical positive one would have been. A claim-decomposition counterfactual analysis separately shows aggregation delivering a real accuracy gain (50.0% vs. 25.0%, $n = 4$) once claim extraction has more than one claim to work with. Two general, ground-truth-independent validator fixes measurably improved recall (16.7%→40%, 100% precision throughout) while leaving reel-level accuracy completely unchanged – a concrete demonstration that verdict-label accuracy and the formally-defined support validity construct (Section IV-D) are separable properties, and, we argue, the more important axis for an architecture whose premise is evidence-before-verdict rather than label-matching. Also reported in full: a gap between finding topically-relevant primary sources (68.75%) and extracting usable evidence from them (18.75%); a multimodal claim-coverage result in which OCR-derived evidence recovered a claim audio

transcription missed entirely; and a structural finding – most of this dataset’s actual false claims live outside the specific post being fact-checked – that sets a real ceiling on single-post claim extraction regardless of component quality. RQ5 (bias) and RQ6 (calibration) are deferred, not forced, because this dataset’s size cannot yet support either – the same discipline held throughout: report what the data shows, including a negative result, a deferred question, a bug in our own baselines, or a threat to validity in our own fix. The taxonomy of concrete failure modes (Appendix A) – 24 entries, over a third found and fixed during this evaluation program itself, all but four with a deterministic check and a regression test built from the case that surfaced it – is as practically useful a contribution as any single number above, since most of these failures, including the baseline confound this conclusion opens with, are only discoverable by running a complete system against real content, not by unit-testing components in isolation. Scaling the benchmark past $n = 9$, closing the claim-extraction coverage gap, and running the matched-pair and calibration analyses this paper withholds are the concrete next steps.

A subsequent foundation-phase program (Section XII) extends this picture without touching the frozen headline comparison above: the benchmark grew to 15 items; two new deterministic validator checks (temporal and entity consistency), each integrated only after independent evaluation cleared a stopping-condition bar, measurably improved the adversarial-benchmark precision and recall (81.8%/50.0%→87.5%/60.9%) with zero new false positives, and directly closed a previously-named, explicitly -disclosed gap case. Most consequentially, four independent experiments – each designed to measure something else, none of them a dedicated reliability study – converged on the same underlying finding: raw local-model output on real content is substantively empty or unusable a large fraction of the time, and this system’s Gemini-escalation defenses are carrying more of the real reliability burden than a component-level pass/fail reading would suggest. Structured 5-query retrieval, a full 8-combination multimodal re-analysis, a candidate 8-category evidence -stance taxonomy, the first stage of a perceptual-hash cross-post detector, and a bounded adversarial stress test were each built and measured with a real, disclosed result – two clear wins, two honest mixed-to-negative findings, and two newly-named, not-yet-investigated structural gaps (input-signal consistency, multilingual extraction bias) – consistent with, not an exception to, the discipline this paper has applied throughout: report what was actually measured, including when it complicates the story.

A second autonomous pass (Section XII-F) closed both named gaps above with real, disclosed outcomes (one infeasible as specified, one corrected rather than confirmed), unlocked the aggregation counterfactual on real VALIDATION data for the first time ($n = 1$, directionally consistent with the original finding), and

extended the adversarial suite into two new categories. A prompt-injection defense with zero prior test coverage was measured live: 1 of 5 attempts against a small local model succeeded, a structural delimiter-spoofing gap was found and fixed independently of model behavior, a hardened prompt reduced the measured failure rate to 0/5, and the downstream anti-hallucination validator was confirmed, not merely assumed, to independently neutralize the one pre-fix success. Separately, a genuinely new failure mode was found and honestly reported as unresolved: verdict-label selection does not reliably weight source reliability under direct contradiction (0 of 14 real trials favored a 0.95-reliability source over a 0.20-reliability, uncited one), and a targeted prompt fix, though kept, measurably failed to resolve it – the third time this program has found prompt engineering insufficient to fully correct a small local model’s behavior, reinforcing rather than contradicting this paper’s central finding that architectural defenses (deterministic checks, Gemini escalation, and now a validator functioning as a measured, not merely claimed, security backstop) carry more of this system’s real reliability burden than prompt design alone.

Appendix

Section IV-D2 summarizes verification-gated model cascading; full detail follows. Five stages – claim extraction, evidence analysis, vision-context, content generation, and verdict proposal – each run a deterministic substantiveness check before accepting local-model output: claim extraction requires at least half of extracted verifiable claims to have a `source_quote` substring-grounded in the transcript/OCR, and at least one claim with non-empty text (zero claims is a legitimate “found nothing,” not a failure); evidence analysis requires a non-empty per-source explanation (firing per-source since this stage calls the LLM per source); vision-context checks `scene_description`’s word-overlap against the system prompt (the 0.20 threshold from Section IV-D2); content generation/verdict reasoning requires at least six real words after stripping citation markup. On failure, the call retries once against a paid model (Gemini) with an identical prompt, logged either way. Illustrative development-time telemetry (predating the frozen evaluation, not part of it) found this gate firing on 3/7 claim-extraction, 2/5 content-generation, and 2/8 verdict-proposal completions – small counts, large enough to show it firing routinely.

Section IV-D3 summarizes deterministic aggregation and grounded corrections; full detail follows. The reel-level rule table: all TRUE yields overall TRUE; a mix of TRUE/MOSTLY_TRUE and FALSE yields MISLEADING. Because a bare FALSE/UNVERIFIED is a weak deliverable, the verdict-proposal call also carries optional `corrected_fact` and `context_note` fields, held to the same evidence-grounding rule as `reasoning_summary`, left null rather than guessed when unestablished. We deliberately did not build the natural next step – speculating about

a claim’s downstream consequences – since that is ungrounded generation of exactly the kind Section IV-D1 exists to prevent. Both fields are invalidated whenever the parent verdict fails validation, even if individually grounded.

Section X-E defines the cross-post attribution problem and states its headline number; full discussion follows. This is structural, not a sampling accident: a fact-checker traces a viral claim back to its original, differently-captioned source video, and that video’s URL is what enters the dataset (Section VI) – a majority of this dataset’s items may be structurally unwinnable for single-post claim extraction regardless of modality coverage. TruthLens has no reverse-video/cross-post matching capability and has never claimed one; we did not prototype a provenance-retrieval module against this program’s own benchmark, since evaluating it on the same 6 items already read would violate our own held-out discipline – a frozen, disjoint pilot set is the correct next step (Section XVI), not attempted here.

Section XIV lists ten threats at a checkable density; full reasoning for each follows, in the same order.

Small, non-independent sample. $n = 9$ (6 with a resolved verdict from every configuration) is far below the 20–30 item target this program set for itself, already a deliberate reduction from an ideal 50–100 (Section V). Every proportion in this paper is reported with an exact count and a Wilson interval so a reader can judge how much these numbers could move with more data – several, including the headline 33.3% TruthLens accuracy figure, have confidence intervals wide enough to be consistent with a genuinely different true rate.

Single domain, single-country content. All evaluated content is Indian political short-form video with English/Hindi/Urdu mixed transcripts. Source-tiering domain lists and research-planning prompts were not tuned for, and have not been tested against, other national contexts.

Single annotator, no inter-annotator agreement. Every draft human judgment used in Sections IX and X (validator TP/FP/TN/FN calls, evidence -relevance judgments, claim-coverage matching) comes from one unadjudicated reviewer on this project. Tier-1 headline verdicts (Section VI) do not have this limitation, since they are independent professional organizations’ own published verdicts, but every finer-grained judgment in this paper does.

The new validator check’s recall improvement is measured on the same sample it was calibrated against. The 40% recall figure (Section IX) is likely optimistic about generalization to differently-phrased “no evidence found” reasoning, since the check’s phrase list was written after reading the two real cases it now catches. The general principle behind the check does not reference ground truth; its specific implementation does, in this narrow sense.

The vision-context fix’s effect on headline accuracy is

not independently re-verified. Section X states this as an expectation (no effect on items 0006/0007, whose root causes are diagnosed as unrelated to vision quality), not a measured result, because the paid fallback’s quota was exhausted before this could be tested end-to-end.

item-0003 is permanently excluded, not scored as a failure. Confirmed unfetchable across three independent attempts spanning this program – a real content-acquisition limitation against Instagram specifically, not addressed by any fix in this paper.

Baseline comparison isolates architecture from model, but not from claim-extraction coverage. All configurations use the same underlying model and search provider (Section V), so a measured accuracy difference is attributable to architecture rather than model quality – but Baselines 1–3 receive the claim as already extracted by TruthLens’s own claim-extraction stage, so the headline comparison cannot separate “TruthLens’s downstream architecture underperforms” from “TruthLens’s own claim extraction gave its downstream architecture nothing to work with for 2 of 6 items,” and Section VII argues explicitly for the latter reading.

The primary-source-retrieval fix is measured on tier classification, not full-text completeness, and on a query sample distinct from Section IX’s main evidence -quality run. The .gov.in full-page-fetch reliability gap (403s, TLS failures) it surfaced remains open, unresolved rather than merely undiscussed.

RQ5 and RQ6 are not evaluated, by design, not by omission. Both are deferred per our own pre-registered fallback rules (Section XI) rather than reported with an underpowered number dressed up as a finding.

Ground truth is itself sourced from a small set of professional fact-checking organizations, whose own editorial judgment we do not independently audit. BOOM Live, Alt News, and Factly are established, widely-cited Indian fact-checking organizations, and using their published verdicts as Tier-1 ground truth is standard practice in this subfield – but this is distinct from RQ5’s question of whether TruthLens treats political actors comparably. A systematic asymmetry in which claims these organizations choose to cover, or how they resolve ambiguous cases, would propagate into every number in this paper undetected, since we have no second, independent ground-truth source to check them against.

Section XI states the headline RQ5/RQ6/efficiency facts; full discussion follows.

RQ5 qualitative observation, in full. Of the 9 real verdicts audited in Section IX, a verdict targeting a government body’s own claim (item-0004’s Delhi Police baton case) was treated no differently in kind by the pipeline than verdicts about party or activist claims – the system was willing to reach FALSE against a claim traceable to a government actor’s own public position (a verdict itself judged wrong for reasons unrelated to which actor was involved). One data point showing the system

does not categorically refuse to find against a government actor.

RQ6 caveated observation, in full. The two highest-confidence verdicts (0.7, 0.8) are both cases Section IX’s draft human review judged unreliable or factually wrong, while several of the lowest-confidence verdicts (0.0–0.1) were appropriately cautious UNVERIFIED outputs. At $n = 9$ this is not evidence of an inverse relationship between confidence and correctness – it is exactly the kind of pattern a real calibration analysis, at real scale, would need to check for rather than assume away.

Efficiency gaps, in full. Baseline 4 (TruthLens minus validation) has an identical call-count/token/latency profile to full TruthLens by construction, since SKIP_VALIDATION changes only which content gets persisted, not which LLM calls get made; no separate timed pass was run for it. Baseline 1 (LLM-only) reuses timing from an earlier script that also did not capture latency, a gap inherited rather than newly introduced here. Exact dollar cost per configuration (Gemini pricing was not itemized per call during this run) is a concrete, scoped gap for a future pass.

This program’s earlier “each configuration’s own full $n = 9$ run” table (Section VII), retained only for the historical record and not cited as evidence for any claim in this paper: 3 of its 9 rows (items 0003/0008/0009) still use the uncorrected claim-input method, since none has both a complete real extraction and a complete real full-TruthLens verdict to correct against (Section VII-A).

TABLE XIV

Superseded: each configuration’s own full run, $n = 9$ (not used for any claim in this paper)

Config	n	Correct	Acc.	95% CI
B1: LLM-only	9	2	22.2%	[6.3,54.7]
B2: Search+LLM	9	7	77.8%	[45.3,93.7]
B3: Search+RAG+LLM	9	5	55.6%	[26.7,81.1]
Full TruthLens	6	2	33.3%	[9.7,70.0]

Section IX summarizes the four-way primary-source metric and three further findings from the evidence-quality audit; full detail follows.

The four novel paraphrase test cases (Check 4 generalization test). “the data does not corroborate this,” “insufficient corroboration exists,” “the available sources do not establish,” and “cannot verify this claim” – none in the original phrase list, each paired with a confident label in a synthetic but realistic evidence matrix; all four were missed.

The four-way primary-source metric, per-metric detail. (1) Source-tier classification rate: $16/68 = 23.5\%$. (2) Relevant primary source rate (draft human judgment): $11/16$ (68.75%) of primary-tier sources are topically relevant; the 5 not relevant include a wrong-domain case (engineering-admissions authority for a medical/NEET claim) and a wrong-country case (US National Archives for an Indian

claim). (3) Primary source fetch-success rate: $16/16$ by construction of an existing guarantee that the pipeline never stores an unfetched source; full-vs-snippet text was not separately checked, a gap shared with the search-side fix below. (4) Usable evidence extraction rate: $3/16 = 18.75\%$ (stance $\neq$ irrelevant).

A second major bug: empty explanations in 47.1% of evidence rows. A real government source was classified contradicts with zero explanation text – no way to audit why. Checking systematically: $32/68$ (47.1%) real evidence rows had an empty explanation field, the same schema-valid-but-empty class found elsewhere, never guarded in this stage. Fixed with a per-source substantiveness check (Section IV-D2); live re-verified against the exact case that surfaced it, producing the same stance with a real, specific explanation. The deeper issue this reveals is less “the stance was wrong” and more that claim extraction had produced a claim general enough that unrelated-event sources become plausible-looking evidence for it – a claim-specificity problem as much as an evidence-relevance one.

Closing the primary-source gap: a deterministic search-side fix. A research-planning query targeted at primary sources is restricted at the search-call level to a fixed government/legal domain allow-list, rather than trusting a free-text site:gov.in the model writes into its own query. A related source-classifier gap (missing NIC-/RBI-hosted domain patterns) was fixed alongside it. Re-running seven real, unscripted domain-restricted queries: four returned results, pooling to 20 sources, 19 of which (95%) landed in a primary tier, against an $\sim 11\%$ ($8/72$) baseline before the fix. Two findings temper this: domain restriction guarantees tier correctness, not topical relevance (one query returned five correctly-tiered but topically irrelevant results, the same pattern as the engineering/NEET and US/Indian mismatches in Section IX); and for one query’s five URLs, every full-page fetch failed (403s, TLS errors), falling back to snippet text – manually confirmed substantive, but this .gov.in fetch-reliability gap remains open, not folded into the 95% headline figure.

A prototype entity-consistency check, evaluated honestly and not integrated. Motivated by Section XIII’s entity-confusion failure mode, we prototyped a deterministic check: does at least one of a claim’s extracted entities appear in each non-irrelevant cited evidence row’s text? Evaluated against the same real 9-claim sample (ENTITY_CONSISTENCY_EVALUATION.md), 7/9 evaluable rows were flagged; manual audit found one genuine true positive (a Delhi -Police claim’s “contradicting” evidence was actually about an unrelated Burdwan, West Bengal incident, refining item-0004’s error attribution), four false positives from a fixable gap (the claim’s only entity was the abstract concept “Democracy,” not a person/organization/location), and one likely false positive from transliteration variance (“Dipke”/“Deepke”). The historical case that most motivated this check – a claim naming no one, whose

evidence smuggled in an unestablished identity – has no entities to check and this design cannot catch it. Not integrated into production: one true positive on ~5 evaluable cases is too small to justify a fifth check without more data (Rule 4). Reported as an honest, mixed result, not suppressed or overclaimed.

Section XIII summarizes this taxonomy; the complete 24-entry list, each with its full description and fix, is given here in full rather than only referenced, per this program’s own discipline against omitting a disclosed finding for space.

- Schema-valid, substantively empty output. Found independently in four different pipeline stages across this project (claim extraction, content generation, verdict reasoning, and, this evaluation program, evidence analysis): a locally hosted model produces output that satisfies its schema – a non-empty string, a well-typed list – while containing no actual information. No schema validation catches this class of failure; only a content-level substantiveness check does (Section IV-D2), and each stage needed its own, since the same architecture existing in one stage did not transfer automatically to the next.
- Verbatim-but-misattributed quotation. A claim’s source quote can be a genuine, verbatim substring of the video’s on-screen text while still being the wrong kind of quote – a caption written by the video’s publisher about a person, not words that person said. Fixed by requiring the candidate speaker’s own name not to appear inside the quote text itself before trusting the attribution.
- Silent verdict/presentation inconsistency. After introducing reel-level, multi-claim verdict aggregation, a review dashboard continued displaying the primary claim’s own individual verdict as “the” verdict, while the actually-published output displayed the separately computed reel-level verdict; the two could and did disagree with no error, because both values were valid data. Found only by comparing rendered output against the database directly.
- Bracket-matching truncation bug. A display-sanitization regex intended to strip an internal validation-note suffix used non-greedy bracket matching, which stopped at the first closing bracket encountered – the closing bracket of a Python list repr inside the note itself – leaving the note’s own tail visible on a published slide. Fixed by truncating at a literal marker string rather than attempting to bracket-match free-form text.
- Unrepresentative thumbnail selection. A single frame extracted at a fixed offset is not a reliable representative thumbnail. Now several candidate frames are sampled and selected by a cheap grayscale-variance “visual complexity” score, screening out blank/flat/transition frames – not real scene- or face-

aware selection, but a materially better floor than one fixed guess.

- Entity confusion between similarly-named organizations. Evidence retrieval for a claim about one organization returned an article about a different, similarly-named one; the evidence-analysis stage labeled this source’s stance as contradicts rather than irrelevant – an article about a different entity does not contradict a claim about this one, it simply does not address it. Not currently checked anywhere in the pipeline.
- Downgraded reasoning reused as trusted input downstream. A verdict-proposal model’s free-text reasoning, containing an entity-confusion-driven fabrication, was correctly downgraded to UNVERIFIED at the label level – but downgrading a verdict forces only its label, not its free-text reasoning summary, because there is no reliable way to isolate the offending sentence from otherwise-reasonable-looking prose. That full text was still being reused verbatim downstream. Fixed by treating any non-passed validation status as disqualifying for reuse of that verdict’s free text anywhere.
- Single-digit fabrication in generated headlines. A headline-generation prompt explicitly instructing the model never to introduce a number not already in the source claim did so anyway, converting “4,000 crore rupees” into an invented, arithmetically wrong “\$1 Billion.” The existing number-grounding check ignores single-digit numbers to avoid flagging incidental meta-commentary in longer prose, an exception that does not apply to a headline this short. Fixed with a headline-specific check requiring every digit sequence in the generated headline to appear in the source claim, falling back to the claim’s own already-grounded text verbatim on failure.
- Vision-read text silently discarded before claim extraction. Vision analysis of a photo correctly read a specific figure and timeframe off an on-screen graphic into a dedicated “visible text” field, distinct from a vaguer scene-description field – but claim extraction’s prompt only ever included the latter. Not a hallucination, but a real signal thrown away before it could be used. Fixed by including the on-screen-text field with the same weight as OCR output.
- Missing glyph rendering as a blank box. The bundled rendering fonts have no glyph for the Indian Rupee sign, a routine character in this domain’s claims, not an edge case. Fixed with a character substitution applied only at render time.
- Vision-transcription accuracy, left unresolved. Distinct from the prompt-echo bug below: the same vision analysis occasionally misreads the text it is looking at (truncated words, a stray character standing in for a digit) without discarding or fabricating it wholesale. This is an accuracy limitation of the underlying vision model itself, not a gap in the

pipeline’s use of it, and we chose not to patch it with an unverified heuristic rather than risk a fix with an unknown false-positive rate.

- Empty-string claims passing schema validation. A locally hosted model returned claims with empty text for content whose OCR plainly contained checkable claims – a non-empty-string type constraint does not require a non-empty string. Fixed with a dedicated substantiveness check plus an unconditional filter that never persists a blank-text claim.
- An except clause written against the wrong SDK exception hierarchy. A real rate-limit response from the fallback provider’s API raised an exception type the provider wrapper’s error handling did not recognize, because the specific API surface in use raises a separate exception hierarchy entirely. This let a routine, expected condition – daily quota exhaustion – propagate as an unhandled exception rather than a clear, typed error. A pre-existing unit test for this exact code path had mocked the same wrong exception hierarchy the production code was mishandling, so it passed throughout – a concrete instance of a test being internally consistent with buggy code while providing zero real coverage of the failure it was written to catch. Fixed by catching the SDK’s actual exception hierarchy and rewriting the test against real instances of it.
- Baseline architecture silently inheriting a rescue mechanism under evaluation. Found while building Baselines 2–3 (Section V): the first working draft called the same provider factory every production stage uses, which silently returns a Gemini-fallback-wrapped provider whenever an API key is set in the environment – giving both “simple” baselines the exact failure-rescue mechanism TruthLens’s own architecture was being evaluated for having, introduced by accident rather than by a bad research decision. Caught during smoke testing before any number was reported anywhere; fixed by importing the local-only provider directly.
- A second, differently-phrased “no video” error masking a fetchable photo post. item-0005 initially failed ingestion outright: yt-dlp reported “No video formats found!” – correct information (it is a photo post) – but the existing photo-post fallback matched only a different, previously-seen phrasing (“no video in this post”). A genuinely fetchable post failed hard instead of falling back to Open Graph metadata. Fixed by matching against a tuple of known phrasings.
- UUID hex fragments leaking through citation markup as fake unsupported numbers. Described in full in Section IX: digits inside a cited evidence ID’s own UUID were being extracted by the number-grounding check as apparent fabricated statistics, in two independently-discovered forms (double- and single-bracket citation markup). The fix’s own side effect – removing an accidental true positive from an earlier validator audit – is itself reported as a finding, not smoothed over.
- 47.1% empty explanations in per-source evidence analysis. Described in full in Section IX: the one LLM-calling pipeline stage with no substantiveness check of its own, confirmed at scale once measured. Fixed with a per-source check.
- Recurring vision-output prompt echo, previously undercounted as a rare anecdote. Described in full in Sections IV-D2 and X: the local vision model repeatedly produces schema-valid, non-empty output that is actually an echo of its own system prompt rather than a real image description, found across at least 4 separate real reels this project – frequent, not rare, once actually counted. Fixed with a word-overlap-with-prompt substantiveness check.
- A label confidently contradicting the model’s own stated reasoning, undetected by any grounding check. Described in full in Section IX: reasoning stating no reliable evidence exists, paired with a confident non-UNVERIFIED label – a distinct failure category from a fabricated citation or number, since every citation and number involved can be entirely real. Fixed with a new deterministic check, with its calibration circularity explicitly disclosed as a threat to validity.
- A MissingGreenlet crash from a stale ORM read after rollback. Found extending a research script to two new items (Section VII): a per-claim exception handler’s db.rollback() expired an already-loaded ORM object, and the next line’s synchronous attribute access on it crashed the entire script instead of recording one claim’s failure – the same bug class already fixed once in production code. Fixed by capturing needed values before the try block.
- A table-generation script’s printed output and its persisted JSON summary silently disagreed, found while building this paper’s figures. day8_final_tables.py reused the same two variable names (k, n) across three separate computations – Table 1’s full-TruthLens row, a per-baseline loop building Table 2, and Table 3’s decomposition ablation – each printed correctly to stdout immediately after being computed, but the summary JSON was assembled at the end of the function from whatever those two names last held, silently combining Table 2’s third baseline’s paired -subset count (n=6, k=3) with Table 3’s item count (n=4) into a fabricated-looking full_truthlens_n6: {n: 4, correct: 3} (75.0%) that appeared nowhere in the script’s own printed tables and did not match Table IV’s real, already-published 33.3% (2/6). This is the same failure class as the silent verdict/presentation-inconsistency entry above – two correct -looking outputs from the same run, one right and one silently wrong – just found in this project’s own research tooling rather than production code, and

exactly why Figure 3 is regenerated from, and checked against, the script’s stdout rather than trusted on the strength of a file simply existing. Caught only because the figure was visually inspected against the table it was supposed to plot before this section was finalized. Fixed by giving every computation its own distinctly-named variables and by persisting Table 2 explicitly (previously only printed), so the artifact a reader or a script can load is no longer a smaller, silently-overlapping namespace than what was shown on screen.

- A quantifiable daily API quota collides with mocked test scenarios, silently, rather than being isolated from them. A quota-aware provider’s global cooldown check (correct, intended behavior: once a real quota exhaustion is recorded, stop attempting real calls against it) reads the same shared table real production usage writes to. Any real exhaustion recorded by concurrent real usage makes mocked test scenarios short-circuit into the real cooldown path before ever reaching their scripted mock response, failing with a missing-row lookup rather than a clear “quota exhausted” message. Diagnosed to the exact free-tier limit responsible: a real 429 response confirmed a 20-requests/day cap on the fallback model used. Not fixed – doing so safely needs either a fully isolated test database or a way to scope the cooldown check to a test context, both larger changes than warranted alongside this finding; the workaround used was re-running the affected tests once concurrent real usage finished, and quantifying the constraint so future runs can anticipate it rather than be surprised by it.
- A structural gap in the delimiter-based prompt-injection defense, and one adversarial style that got past it. The convention every LLM-calling stage uses to mark reel-derived text as data, not instructions, never neutralized literal occurrences of its own delimiter tokens already present in the untrusted text being wrapped – a transcript or caption containing the literal closing token could, in principle, forge a fake data-block boundary regardless of how reliably any given model resists in-band instructions. Fixed deterministically (sanitizing those tokens before wrapping) and verified in both directions (reintroducing the bug on a scratch copy confirmed a new regression test catches it; restoring the fix confirmed it passes again). Separately, of 5 live adversarial prompt-injection attempts against claim extraction, the most direct style (a blunt “ignore all previous instructions” override) succeeded once, producing the injected false claim verbatim as a verifiable, maximum-importance claim – confirming, for the first time with a real measurement rather than an assertion, this project’s own prior documented expectation that small local models follow in-data instructions more readily than larger ones. A hardened extraction prompt reduced

this to 0/5 on re-test; carrying the one pre-fix success through the real remainder of the pipeline showed the downstream anti-hallucination validator independently resolving it to a correct, evidence-grounded FALSE verdict – a real, if single-case, confirmation that the validator functions as the claimed backstop even when an earlier defense is bypassed.

- Verdict-label selection does not reliably weight source reliability under direct contradiction. Constructed evidence with a 0.95-reliability primary-government source supporting a claim and a 0.20-reliability, uncited source contradicting it produced TRUE or MOSTLY_TRUE in 0 of 14 real trials across two prompt versions – including runs whose own free-text reasoning correctly identified the contradicting source as unreliable, yet still did not choose a label reflecting that assessment; one run paired an incorrect label with maximum (1.0) stated confidence. A prompt hardened with an explicit reliability-weighting rule and worked example did not resolve this on re-test (still 0/7) – reported as a measured negative result, not silently dropped, with a deterministic validator check (comparing cited-evidence reliability against verdict direction) named as the likely real fix and left for a dedicated future pass rather than added without its own evaluation.

Full numeric detail for Section XII, one subsection per experiment; every number traces to experiments/registry.jsonl and a dated results artifact under research/results/.

A. EXP-013/EXP-019: validator checks, full numbers

Check 6 (temporal) and Check 7 (entity) were each unit-tested against real observed values before integration (7 and 7 regression tests respectively), then jointly stress-tested on the expanded 34-case adversarial benchmark. Combined confusion counts: TP=14, FP=2, FN=9, TN=9 (precision 87.5%, recall 60.9%). Both false positives are valid-9/valid-10, a pre-existing Check-3 limitation (a contrastive “the true figure is X, not the claimed Y” reasoning pattern false-positives because Y, the refuted number, is never present in the evidence passage – only the claim text contains it, and Check 3 only has access to evidence passages). A live end-to-end verification attempt (EXP-013) against a real claim with a resolvable time_reference was inconclusive by circumstance – an earlier check (missing citation) fired first because the real search results for that specific claim were mostly irrelevant – disclosed rather than re-run at further real cost, since Check 6’s logic is independently covered by direct unit tests.

B. EXP-012/EXP-016: source-quote groundedness

EXP-012 measured whether claim_extraction_confidence correlates with claim groundedness (a verbatim-quote match against the item’s real content) across 8 real DEV-split items: 12 real, substantive,

non-empty claims, 0 grounded – zero variance blocked the intended correlation computation entirely. EXP-016 independently re-measured the same groundedness signal at much higher n (90 real claims, 8 items $\times$ 8 modality combinations): also 0 grounded. Spot-checked directly, not assumed a measurement artifact: claim text is real and substantive (e.g., “#CJP protesters allegedly pelted stones at a new hydrogen train”); the model simply does not populate `source_quote` in practice, regardless of which input signals it is given.

C. EXP-016: full 8-combination coverage

Ground-truth claim coverage (same 8 real items as Section X, manual matching against `items.jsonl`): `caption_only` 25.0%, `audio` 25.0%, `ocr` 25.0%, `vision` 37.5%, `audio+ocr` 25.0%, `audio+vision` 25.0%, `ocr+vision` 12.5%, `audio+ocr+vision` 25.0%. Two items produced zero claims in every one of the 8 conditions, including, notably, this dataset’s one TRUE-labeled item. A third item produced 5 real claims under the richest condition, none of which covered the actual ground-truth assertion – a genuine wrong-claim-extracted failure (Rule 5), distinct from the zero-claims failure mode.

D. EXP-015: 5-query retrieval, full numbers

4 real DEV claims, 20 queries (5 types $\times$ 4 claims), 42 sources fetched, 16 primary-tier. Per-query-type relevant-primary-source rate (manually judged): `primary_source` 3/4 (75%), `context_history` 2/2 (100%, $n = 2$), `contradiction` 1/3 (33%), `entity_focused` 1/4 (25%), `exact_claim` 0/3 (0%) – the last driven entirely by one claim (a private organization with no natural government angle) whose 7 primary-tier hits were all keyword-collision false matches on unrelated `.gov.in/.nic.in` domains (a Gujarat seismology institute’s insurance-claims page, an income-tax assessment page, a Supreme Court hearing list, among others) – exposing that this system’s domain-pattern tier classification has no content-relevance check of its own.

E. EXP-017: evidence-stance taxonomy, full numbers

An 8-category candidate stance taxonomy (4 new categories: `same_event_wrong_entity`, `temporally_mismatched`, `insufficient_detail`, `mentions_only`) was compared against 20 real, already-persisted evidence rows. Raw category-change rate was 55% (11/20), but 80% (16/20) of rows had an empty explanation on at least one side because this specific comparison did not replicate the production substantiveness-retry safeguard – filtered to the 4 rows where both sides were genuinely substantive, only 1 changed category, too small an n for any real conclusion about the taxonomy itself. Not integrated; the comparison’s own confound is the more important, and more general, finding (Section XII-C).

F. EXP-018: cross-post detection, full numbers

Same-video positive control (one real reel, frames independently re-extracted twice via separate `ffmpeg` sampling passes): 25 vs. 25 frames, minimum Hamming distance 0. Different-video negative control (two different real reels): 25 vs. 21 frames, minimum Hamming distance 20 (match threshold 10). A genuine two-independently-posted-same-footage pair (this dataset’s own item-0002, whose ground-truth notes name a specific real original source) was searched for live but its exact post URL could not be extracted from the citing fact-check article through available tooling – disclosed as not achieved this pass rather than forced or approximated.

G. EXP-020: adversarial stress test, full numbers

Six synthetic, unpersisted cases run through the real, unmodified claim -extraction stage: `garbled_ocr` (0 claims), `caption_transcript_contradiction` (1 claim, from the caption only – the unrelated transcript content was silently dropped, not merged), `extremely_short` (0 claims), `mixed_language_chaos` (2 claims, both from the English-script portion of a Hindi/Urdu/English sentence), `repetitive_spam` (40 repetitions of one sentence collapsed to 1 claim), `near_max_tokens` (0 claims, confounded by a real Gemini quota-exhaustion event during the one quality-retry this case triggered). Zero crashes across all six.

H. EXP-021: source_quote prompt strengthening, full numbers

A candidate prompt (`claim_extraction.v4-quote-candidate`) added an explicit 4-step self-check procedure and a worked example matching the real “5,154 likes” OCR case, identical otherwise to production’s v3. Re-run against the same 8 real DEV items as EXP-012/EXP-016: baseline (v3, re-run fresh) produced 9 claims, 0 with `source_quote` (0.0%); the candidate (v4) produced 6 claims, 0 with `source_quote` (0.0%). On the one item containing the literal “5,154 likes” text the worked example was built from, v3 produced 7 unquoted claims and v4 produced zero claims at all – the stronger instruction did not recover the quote and may have suppressed extraction on that item entirely.

I. EXP-022: input-signal consistency feasibility, full numbers

Jaccard similarity of stopword-filtered keywords between `caption_text` and `transcript`, computed for every real benchmark reel with both fields populated ($n = 4$): 0.000 in all four cases (zero shared keywords). Not a sign of inconsistency – the normal case for this project’s real, legitimate, multilingual content.

J. EXP-023: multilingual bias re-examination, full numbers

Script classification (Latin vs. Devanagari/Urdu Unicode ranges) of all 90 real claims from EXP-016’s own

results file (no new extraction, pure re-analysis). Overall: 42 Latin, 48 non-Latin. Per-item: a fully-Marathi item produced 37 non-Latin vs. 8 Latin claims (the Latin ones being caption metadata – like/comment counts – not content); a fully-Hindi item (Karni Sena) produced 5/5 non-Latin; two English-language items produced Latin-only claims; and the one genuinely code-switched real item (Hindi-English mixed) produced claims in both scripts (7 Latin, 6 non-Latin) – direct evidence against a general Latin-script bias.

K. EXP-024: VALIDATION-split real population, full numbers

The real, unmodified second half of `orchestrator.analyze_reel()` (claim extraction through verdict) was run and committed for real against all 6 VALIDATION-split items. item-0011/0012/0013/0015 produced zero claims. item-0010 produced 2 claims (one downgraded entity_mismatch, one FALSE/passed); item-0014 produced 1 claim (FALSE/passed, matching real ground truth). The entity-mismatch case was manually audited against its cited source: the claim (“a female student was severely beaten,” in the context of a Jharkhand-police protest item) was cited against a real news article about an unrelated 2026 IIT Palakkad security incident – a genuine, correct catch, confirming Check 7 generalizes to real production data it was never calibrated against.

L. EXP-025: aggregation counterfactual re-run on real VALIDATION data, full numbers

Filtering the 6 VALIDATION items to those with ≥ 2 verifiable claims each carrying a real, current verdict yields exactly $n = 1$ (item-0010; real ground truth FALSE). Single-claim-only (the highest-importance claim’s own verdict, itself UNVERIFIED after Check 7’s entity-mismatch downgrade) misses. Full decomposition via the real, unmodified `derive_overall_verdict()`, combining both claims, correctly resolves to FALSE. Directionally consistent with, not statistical confirmation of, the original $n = 4$ DEV-split result.

M. EXP-026/EXP-027: prompt-injection stress test, full numbers

5 synthetic Reel cases, each embedding a differently-styled injection attempt in a different field (direct override in transcript; forged delimiter tokens plus a fake system message in caption; a roleplay/DAN jailbreak in OCR; an appeal-to-authority override in transcript; an override buried inside otherwise-legitimate content), run through the real, unmodified `claim_extraction.extract_claims()`. Pre-fix (`claim_extraction.v3`): 1/5 succeeded (the direct-override case). Post-fix (delimiter sanitization plus a hardened `claim_extraction.v4` prompt): 0/5. EXP-027 carried the one pre-fix success through the real remainder of the pipeline: 12 sources fetched, 12 evidence rows produced, final verdict FALSE, `validation_status=passed`, reasoning

grounded in real cited evidence (e.g. citing the absence of documented radiation-linked illness among Apollo astronauts, contradicting the injected moon-landing-hoax premise).

N. EXP-028: evidence-stance taxonomy comparison re-run, full numbers

Identical to EXP-017 except the new-taxonomy call now runs the same `_explanation_looks_substantive` check plus Gemini quality -retry production’s `evidence_analysis.py` itself uses, against the same 20 real Evidence rows. Attempt 1: every retry attempt hit an already-active Gemini cooldown from concurrent earlier work (0/20 retries fired); clean rows 6/20. Attempt 2, run after that cooldown’s own end time: the first retry attempt returned a real HTTP 429 (“Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 20”), setting a fresh cooldown that blocked the remaining 19 retries too; clean rows 5/20. One clean reclassification (contradicts $\rightarrow$ temporally_mismatched) is reported but assessed as a likely taxonomy-misapplication (the source’s substantive numerical disagreement is what actually matters, not that its rules happened to be dated), not cherry-picked as a positive result.

O. EXP-029: contradictory-sources verdict stress test, full numbers

3 synthetic cases (constructed Claim/Evidence/Source rows with explicit, differing reliability scores) run through the real, unmodified `verdict.propose_verdict()`. The `reliability_weighted_conflict` case (0.95-reliability primary-government support vs. 0.20-reliability uncited contradiction) was re-run repeatedly to characterize stability: pre-fix ($n = 7$, `verdict.v2`) produced MOSTLY_FALSE $\times 3$ (confidence 0.0), UNVERIFIED $\times 3$ (confidence 0.4), OUTDATED $\times 1$ (confidence 1.0); post-fix ($n = 7$, `verdict.v3`, after an explicit reliability-weighting rule was added to the prompt) produced UNVERIFIED $\times 5$, MOSTLY_FALSE $\times 2$, confidence 0.0 throughout. 0 of 14 total runs produced TRUE/MOSTLY_TRUE. A sanity check on the `majority_with_credible_outlier` case (3 established -news sources agree, 1 higher-reliability outlier disagrees) showed similar post-fix instability (2/4 MOSTLY_TRUE, 2/4 UNVERIFIED), though with only 1 pre-fix sample a clean before/after comparison isn’t possible for that case.

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